

A linear bound for the Paulsen problem

Astra* Claude† Kaarel Hänni Hugo Eberhard

October 2026

Abstract

We prove that every real ε -nearly equal-norm Parseval frame of n vectors in $\mathbb{R}^d$ is within squared Frobenius distance $C\varepsilon d$ of an equal-norm Parseval frame, where C is a universal constant. This is optimal up to the value of C . Equivalently, every rank- d orthogonal projection on $\mathbb{R}^n$ whose diagonal deviates from d/n by at most β is within squared Frobenius distance $Cn\beta$ of a rank- d projection with constant diagonal. The proof combines a deterministic *static balancing theorem*—a row scaling, obtained by minimising a log-determinant over a box, that corrects any diagonal error which is small compared with the inverse of a half-set hitting time (“barrier constant”) of the random walk with rates P_{ij}^2 —with two random *seeds* that make this hitting time small at linear cost: a filtered Gaussian tangent perturbation with a deterministic drift when $d \gtrsim \log n$, and a row-wise tangent perturbation with row renormalisation when $n \gtrsim d^2$. Bounded d is handled by the Hamilton–Moitra argument. The main theorem has been formalised in Lean 4 with Mathlib.

Contents

1	The result and the plan	2
1.1	Statement	2
1.2	The plan	3
2	The scaling toolbox	4
2.1	Projections, distances, and the squared-Gram Laplacian	4
2.2	Diagonal scaling	5
2.3	The barrier constant and static balancing	6
2.4	Barriers from a dense core	7
2.5	From a seed to an exact frame	8
3	Bounded rank: the Hamilton–Moitra estimate	8
4	The many-row seed	9
4.1	The noise	9
4.2	Row normalisation and the exact second-order identity	10
4.3	A dense reference graph	11
4.4	Proof of Theorem 4.1	11

*AI system (gpt-astra).

†AI system, Anthropic.

5	The moderate-row seed	12
5.1	The horizontal tangent space and the filtered covariance	13
5.2	Commutator identities	14
5.3	The second-order mean and the drift	14
5.4	The expected graph	15
5.5	A good sample	15
5.6	The drifted retraction	17
5.7	Proof of Theorem 5.1	18
6	Assembly	18
A	Gaussian tools	19
B	Formalisation	20

1 The result and the plan

1.1 Statement

A frame of n vectors in $\mathbb{R}^d$ is a matrix $X \in \mathbb{R}^{n \times d}$ whose i th row is x_i^T . Distances are *squared* Frobenius distances $\|X - Y\|_F^2 = \sum_i \|x_i - y_i\|^2$, with no normalisation. Throughout,

$$a = d/n.$$

X is *Parseval* if $X^T X = I_d$ and *equal-norm* if $\|x_i\|^2 = a$ for all i ; an equal-norm Parseval frame is an *ENP frame*. X is an ε -*nearly ENP frame* if

$$(1 - \varepsilon)I_d \preceq X^T X \preceq (1 + \varepsilon)I_d, \quad (1 - \varepsilon)a \leq \|x_i\|^2 \leq (1 + \varepsilon)a \quad (1 \leq i \leq n). \quad (1.1)$$

Theorem 1.1 (Linear Paulsen bound). *There is an absolute constant C such that for all $1 \leq d \leq n$, all $0 < \varepsilon < 1$ and every real ε -nearly ENP frame $X \in \mathbb{R}^{n \times d}$ there is an ENP frame $W \in \mathbb{R}^{n \times d}$ with $\|X - W\|_F^2 \leq C\varepsilon d$.*

The bound is optimal up to the constant (Remark 1.3). Kwok, Lau, Lee and Ramachandran [3] proved the first bound independent of n , $O(\varepsilon d^{13/2})$; Hamilton and Moitra [2] improved it to $O(\varepsilon d^2)$; Lau and Ramachandran announced the linear bound [4], via a different route. Appendix B describes a Lean 4 formalisation of Theorems 1.1 and 1.2.

For a Parseval U write $P = UU^T$ for the orthogonal projection onto its column space; then $P_{ii} = \|u_i\|^2$. The problem has an equivalent form in which the error is additive.

Theorem 1.2 (Projection form). *There is an absolute constant C such that for every $n \geq 1$ and every rank- d orthogonal projection P on $\mathbb{R}^n$, with $\beta = \max_i |P_{ii} - d/n|$, there is a rank- d orthogonal projection Q with $Q_{ii} = d/n$ for all i and $\|P - Q\|_F^2 \leq Cn\beta$.*

This reformulation and the complement symmetry are due to Cahill and Casazza [1]: both β and $\|P - Q\|_F$ are unchanged when (P, Q) is replaced by $(I - P, I - Q)$, so in Theorem 1.2 one may assume $d \leq n/2$. In Section 6, Theorem 1.2 is derived from the main proposition and Theorem 1.1 from Theorem 1.2.

Remark 1.3 (Optimality). Let $d = 2d_1$, $n = 2n_1$, and $1 \leq k \leq n_1/4$ with $n_1 - k \geq d_1$; put $\varepsilon = 4k/n$. Let $X = X_1 \oplus X_2$, where X_1 is an ENP frame of $n_1 - k$ vectors in $\mathbb{R}^{d_1}$ and X_2 one of $n_1 + k$ vectors in $\mathbb{R}^{d_1}$ (Corollary 3.2). Then X is Parseval, and its squared row norms $a(1 \mp 2k/n)^{-1}$ make it an ε -nearly ENP frame. Let W be ENP, and let E and A be the principal blocks of XX^T and WW^T on the first $n_1 - k$ rows. Then E is a projection of trace d_1 and A a positive contraction of trace $(n_1 - k)a$, so $\Delta = E - A$ has $\text{tr } \Delta = ka = \varepsilon d/4$. Since $\text{tr}(E\Delta) = \text{tr } \Delta + \text{tr}((I - E)A) \geq \text{tr } \Delta$,

$$\text{tr}(A - A^2) = -\text{tr } \Delta + 2\text{tr}(E\Delta) - \text{tr } \Delta^2 \geq \text{tr } \Delta - \|\Delta\|_F^2.$$

As WW^T is a projection, its off-diagonal block has squared norm $\text{tr}(A - A^2)$, while that of XX^T vanishes; hence $\|XX^T - WW^T\|_F^2 \geq \|\Delta\|_F^2 + 2(\text{tr } \Delta - \|\Delta\|_F^2) \geq 2\text{tr } \Delta - \|XX^T - WW^T\|_F^2$, i.e. $\|XX^T - WW^T\|_F^2 \geq \text{tr } \Delta$, and by Lemma 2.1 $\|X - W\|_F^2 \geq \frac{1}{2}\text{tr } \Delta = \varepsilon d/8$.

Conventions. c, C denote positive absolute constants whose value may change between occurrences; constants with subscripts or names are fixed once chosen. $\|\cdot\|_{\text{op}}$ is the Euclidean operator norm, $\mathbf{1}$ the all-ones vector, $\text{Diag}(x)$ the diagonal matrix with diagonal x , $\text{diag}(M)$ the diagonal of M as a vector, and $\Pi_{\mathcal{S}}$ the orthogonal projection onto a subspace $\mathcal{S}$. $A \preceq B$ is the Loewner order. All Gaussian vectors are real and centred.

1.2 The plan

The engine: scaling. Let U be Parseval with projection P and diagonal $p = \text{diag } P$. For $s \in \mathbb{R}^n$ the matrix $\text{Diag}(e^s)U$ has a column space with projection $P(s)$, and $s \mapsto \text{diag } P(s)$ is the gradient of the convex function $\frac{1}{2} \log \det(U^T \text{Diag}(e^{2s})U)$, whose Hessian at $s = 0$ is $2L_P$, where

$$L_P = \text{Diag}(p) - P \circ P$$

is the graph Laplacian with edge weights P_{ij}^2 (Lemma 2.2). To first order, then, changing the diagonal by $\delta \perp \mathbf{1}$ requires $s \approx \frac{1}{2}L_P^+ \delta$, and the cost of the move is about $s^T L_P s \approx \frac{1}{4}\delta^T L_P^+ \delta$. If L_P has spectral gap λ and $|\delta_i| \leq \varepsilon a$, the cost is at most $n\varepsilon^2 a^2/\lambda$. For a “random-looking” P one has $\lambda \approx a$ and the cost is $\varepsilon^2 d$, but L_P can be arbitrarily badly connected: for an exact direct sum the components cannot exchange diagonal mass at all.

Perturb, then scale. Perturb U by a random horizontal tangent of size $t = \sqrt{K_0\varepsilon}$, at cost $O(t^2 d) = O(\varepsilon d)$. Generic entries of the new projection then have size $t\sqrt{a/n}$, so most edges acquire weight $\gtrsim at^2/n$, and the random walk with rates P_{ij}^2 reaches any half of the vertices in time $O(1/(at^2))$: the *barrier constant* (Definition 2.6 and Lemma 2.10) is $O(1/(at^2))$. The static balancing theorem (Theorem 2.8) then removes any diagonal error β with $\beta \cdot O(1/(at^2)) \leq \frac{1}{2}$ exactly, at cost $\frac{15}{4}n\beta = O(t^2 d)$. The ℓ_∞ (rather than spectral) formulation is essential: the scaling must stay in a box $\|s\|_\infty \leq 1$, and an ℓ_2 argument loses a factor $\sqrt{n}$.

The difficulty. So the perturbed frame must have diagonal error $\leq \theta at^2$ in ℓ_∞ for a small fixed θ . The initial error $\varepsilon a = at^2/K_0$ is fine. But a random tangent of size t moves each diagonal entry *linearly* by about $t\sqrt{a/n}$, which exceeds at^2 as soon as $\varepsilon \lesssim 1/d$. Two different constructions avoid this.

Seed 1: many rows ($n \geq Bd^2$, [Section 4](#)). Perturb each row *orthogonally to itself* and renormalise. Row norms stay exact; the cost is that the frame operator is no longer exactly I . Its first order change is removed by conditioning the Gaussian on a subspace of codimension $\leq d^2$, and its second-order fluctuation is an average over n rows, of size $t^2 d / \sqrt{n}$. The spectral error of the perturbed frame is therefore $\eta + O(t^2 d / \sqrt{n} + t^2 d^2 / n + t^3)$, with t^2 a large multiple of η , which is at most θt^2 once that multiple and B are large and t is small; whitening turns a spectral error δ into a diagonal error at most $2a\delta$. The cubic term is $O(t^3)$ rather than the termwise $O(t^3 d)$ because the first-order constraint lets one centre the normalisation weights ([Lemma 4.4](#)); this is what makes the seed work for every $\eta \leq c_*$, with no further restriction relating η and d .

Seed 2: moderate rows ($d \geq A \log 2n$, [Section 5](#)). Perturb in the horizontal tangent space of the Grassmannian (so the frame stays exactly Parseval) with a Gaussian whose covariance is filtered: $C_\rho = \rho(I - \Omega)(\rho I + \Omega)^{-1}$, where Ω is the normalised Gram operator of the linear diagonal map $\mathcal{A}$. The filter makes the linear diagonal change $O(t\sqrt{\rho a \log n/n})$, negligible once $\rho \asymp t^2$ and $d \gg \log n$, while leaving weak (connecting) directions untouched. The price is a second-order bias $t^2 b_*$, which is not small in ℓ_∞ . A commutator identity shows that $b_* = \mathcal{A}m_*$ for an explicit deterministic tangent direction m_* with $\|m_*\|_F^2 \leq 2a/\rho$ ([Lemmas 5.5 and 5.6](#)). We simply add the drift $\frac{t}{2}m_*$ to the noise: its first-order effect cancels the bias exactly, and its other effects are $O(at^2/D)$. All remaining errors are $\leq \theta at^2$ by concentration, which is where $d \geq A \log 2n$ is used. Only a bounded, deterministic set of rows needs a spectral argument, and second moments suffice there.

Assembly ([Section 6](#)). After complementation and normalisation it suffices to treat equal-norm frames with $2d \leq n$ and spectral error η . If $d < d_0$ (a large absolute constant), the Hamilton–Moitra bound $\eta d(d-1) \leq d_0 \eta d$ suffices ([Section 3](#)). If $d \geq d_0$ and $n \geq Bd^2$, use Seed 1. If $d \geq d_0$ and $n < Bd^2$, then $A \log 2n \leq A \log(2Bd^2) \leq d$ and Seed 2 applies.

regime	construction	where
$d < d_0$	radial isotropic position, ordered coordinates (Hamilton–Moitra)	Section 3
$d \geq d_0, n \geq Bd^2$	row-tangent Gaussian, row normalisation, whitening, balancing	Section 4
$d \geq d_0, 2d \leq n < Bd^2$	filtered horizontal Gaussian with drift, balancing	Section 5
$d > n/2$	complement	Section 6

2 The scaling toolbox

Throughout this section $U \in \mathbb{R}^{n \times d}$ is Parseval, $P = UU^T$, $p = \text{diag } P$, and $L = L_P = \text{Diag}(p) - P \circ P$. For nonnegative symmetric weights w_{ij} the Laplacian is $(Lx)_i = \sum_j w_{ij}(x_i - x_j)$; diagonal weights do not contribute. We write $\text{osc}(x) = \max_i x_i - \min_i x_i$.

2.1 Projections, distances, and the squared-Gram Laplacian

Lemma 2.1 (Frames versus projections). *Let $U, W \in \mathbb{R}^{n \times d}$ be Parseval with projections P, Q , and let $\sigma_1, \dots, \sigma_d \in [0, 1]$ be the singular values of $U^T W$. Then*

$$\min_{R \in O(d)} \|U - WR\|_F^2 = 2 \sum_k (1 - \sigma_k) \leq \|P - Q\|_F^2 = 2 \sum_k (1 - \sigma_k^2) = 2\|(I - P)W\|_F^2 \leq 2\|U - W\|_F^2.$$

Right multiplication by $R \in O(d)$ preserves Parsevalness and every row norm. If $X \in \mathbb{R}^{n \times d}$ has $\delta = \|X^T X - I\|_{\text{op}} < 1$, its polar factor $\tilde{X} = X(X^T X)^{-1/2}$ is Parseval, $\|X - \tilde{X}\|_F^2 \leq d\delta^2$, and

$\|x_i\|^2/(1+\delta) \leq \|\tilde{x}_i\|^2 \leq \|x_i\|^2/(1-\delta)$. If moreover $\delta \leq \frac{1}{2}$, then $|\|\tilde{x}_i\|^2 - \|x_i\|^2| \leq 2\delta\|x_i\|^2$ and row i of $\tilde{X}\tilde{X}^T - XX^T$ has squared norm at most $6\delta^2\|x_i\|^2$.

Proof. $\|U - WR\|_F^2 = 2d - 2\text{tr}(U^T WR)$, maximised over R by the polar factor of $U^T W$, giving $2d - 2\sum_k \sigma_k$. Next $\|P - Q\|_F^2 = 2d - 2\text{tr}(PQ) = 2d - 2\|U^T W\|_F^2$ and $\|(I - P)W\|_F^2 = d - \|U^T W\|_F^2$; also $1 - \sigma \leq 1 - \sigma^2$ on $[0, 1]$, and $(I - P)W = (I - P)(W - U)$ gives $\|(I - P)W\|_F \leq \|U - W\|_F$. For the polar factor, if ς_k are the singular values of X then $\|X - \tilde{X}\|_F^2 = \sum_k (\varsigma_k - 1)^2 \leq \sum_k (\varsigma_k^2 - 1)^2$, and $\|\tilde{x}_i\|^2 = x_i^T (X^T X)^{-1} x_i$. With $E = (X^T X)^{-1} - I$, so $\|E\|_{\text{op}} \leq \delta/(1 - \delta) \leq 2\delta$, row i of $\tilde{X}\tilde{X}^T - XX^T = XEX^T$ is XEx_i , of squared norm at most $(1 + \delta)4\delta^2\|x_i\|^2$. $\square$

Lemma 2.2 (Energy identity). L is the Laplacian of the weights P_{ij}^2 ($i \neq j$), $L \succeq 0$, $L\mathbf{1} = 0$, $L_{I-P} = L_P$, and for $x \in \mathbb{R}^n$, $X = \text{Diag}(x)$,

$$x^T Lx = \frac{1}{2} \sum_{i,j} P_{ij}^2 (x_i - x_j)^2 = \|(I - P)XU\|_F^2 = \frac{1}{2} \| [X, P] \|_F^2.$$

Proof. $\sum_j P_{ij}^2 = (P^2)_{ii} = p_i$, so the rows of $\text{Diag}(p) - P \circ P$ sum to zero and its off-diagonal entries are $-P_{ij}^2$. Next $\|(I - P)XU\|_F^2 = \text{tr}(XPX(I - P)) = \sum_i x_i^2 p_i - \sum_{i,j} x_i x_j P_{ij}^2$, and $[X, P]_{ij} = (x_i - x_j)P_{ij}$. For $I - P$ the off-diagonal weights are again P_{ij}^2 . $\square$

Lemma 2.3 (Scaled projections). For $w \in \mathbb{R}_{>0}^n$ put $D = \text{Diag}(w)$, $M = (U^T D^2 U)^{-1/2}$ and $W = DUM$. Then W is Parseval with column space $D \text{im } U$, and

$$\|WW^T - P\|_F^2 = 2\|(I - P)W\|_F^2 \leq \frac{2w^T Lw}{\min_i w_i^2}.$$

Proof. $W^T W = MU^T D^2 U M = I$ and $U^T D^2 U \succeq (\min_i w_i)^2 I$, so $\|M\|_{\text{op}} \leq 1/\min_i w_i$. By Lemmas 2.1 and 2.2, $\|(I - P)W\|_F^2 \leq \|M\|_{\text{op}}^2 \|(I - P)DU\|_F^2 = \|M\|_{\text{op}}^2 w^T Lw$. $\square$

2.2 Diagonal scaling

For $s \in \mathbb{R}^n$ let $D_s = \text{Diag}(e^s)$, let $P(s)$ be the projection onto $D_s \text{im } U$, $r(s) = \text{diag } P(s)$, and $F(s) = \frac{1}{2} \log \det(U^T D_s^2 U)$.

Lemma 2.4. F is smooth and convex, $\nabla F(s) = r(s)$, $\sum_i r_i(s) = d$, and $F(s + c\mathbf{1}) = F(s) + cd$.

Proof. With $G = U^T D_s^2 U = \sum_i e^{2s_i} u_i u_i^T$ one has $\partial_i F = e^{2s_i} u_i^T G^{-1} u_i = P(s)_{ii}$, because $P(s) = D_s U G^{-1} U^T D_s$; differentiating again gives $\nabla^2 F = 2L_{P(s)} \succeq 0$. The trace of $P(s)$ is d , and G scales by e^{2c} under $s \mapsto s + c\mathbf{1}$. $\square$

Lemma 2.5 (One-sided scaling inequalities). For $z \in \mathbb{R}_{>0}^n$ let r be the diagonal of the projection onto $\text{Diag}(\sqrt{z}) \text{im } U$. Then, coordinatewise,

$$Lz \leq z \circ (r - p), \quad L(z^{-1}) \leq -z^{-1} \circ (r - p).$$

Proof. Let $G = U^T \text{Diag}(z)U$. Since $G + z_i^2 G^{-1} - 2z_i I = (G^{1/2} - z_i G^{-1/2})^2 \succeq 0$, evaluating at u_i and using $u_i^T G u_i = \sum_j P_{ij}^2 z_j = p_i z_i - (Lz)_i$ and $z_i^2 u_i^T G^{-1} u_i = z_i r_i$ gives the first inequality. Let $V \in \mathbb{R}^{n \times (n-d)}$ be an isometry onto $(\text{im } U)^\perp$. The subspaces $\text{Diag}(\sqrt{z}) \text{im } U$ and $\text{Diag}(z^{-1/2}) \text{im } V$ are orthogonal complements, so the projection onto the latter has diagonal $\mathbf{1} - r$, while $VV^T = I - P$ has diagonal $\mathbf{1} - p$ and Laplacian L (Lemma 2.2). The first inequality for (V, z^{-1}) is the second. (If $d = n$ then $L = 0$ and $r = p = \mathbf{1}$.) $\square$

2.3 The barrier constant and static balancing

Definition 2.6 (Barrier constant). Let L be a weighted Laplacian on $[n]$. For $S \subseteq [n]$, an S -barrier is a vector $\psi \geq 0$ with $(L\psi)_i \geq 1$ for every $i \notin S$. The *barrier constant* $H(L) \in (0, \infty]$ is the least H such that every S with $|S| \geq n/2$ has an S -barrier with $\|\psi\|_\infty \leq H$ ($H(L) = \infty$ if some such S has no barrier).

For the continuous-time random walk that jumps from i to j at rate w_{ij} , the expected hitting time $h(x) = \mathbb{E}_x \tau_S$ vanishes on S and satisfies $(Lh)_i = 1$ off S ; and if ψ is any S -barrier and $\epsilon > 0$, the maximum principle argument in the proof of [Lemma 2.7](#) applied to $h - (1 + \epsilon)\psi$ gives $h \leq (1 + \epsilon)\psi$. So h is the minimal S -barrier, and $H(L_P)$ is the half-set hitting time of the walk with rates P_{ij}^2 . If $H(L) < \infty$ then the graph is connected: for a component C with $|C| \leq n/2$, summing $(L\psi)_i$ over C gives 0, so C^c has no barrier. It is within a factor 2 of the ℓ_∞ Poisson constant of L ([Remark 2.11](#)). A small barrier constant controls scalings in ℓ_∞ :

Lemma 2.7 (Median barrier). *Let $H(L) \leq H < \infty$, let $\beta \geq 0$ with $\beta H < 1$, let $z \in \mathbb{R}_{>0}^n$, and let m be a median of z (so that $\{z \leq m\}$ and $\{z \geq m\}$ both have at least $n/2$ elements). If*

$$(Lz)_i \leq \beta z_i \text{ when } z_i > m, \quad (Lz^{-1})_i \leq \beta z_i^{-1} \text{ when } z_i < m,$$

then $(1 - \beta H) \max_i z_i \leq m \leq (1 - \beta H)^{-1} \min_i z_i$.

Proof. Let $S = \{z \leq m\}$, let ψ be an S -barrier with $\|\psi\|_\infty \leq H$, let $M = \max z$ and $\beta' > \beta$. The vector $v = z - m\mathbf{1} - \beta' M\psi$ is ≤ 0 on S , and for $i \notin S$, $(Lv)_i \leq \beta z_i - \beta' M < 0$. At a maximiser i of v , $(Lv)_i = \sum_j w_{ij}(v_i - v_j) \geq 0$; so every maximiser lies in S , and $v \leq 0$. Hence $M \leq m + \beta' MH$; let $\beta' \downarrow \beta$. Apply the same argument to z^{-1} , whose median is m^{-1} , with $S = \{z \geq m\}$. $\square$

Theorem 2.8 (Static balancing). *Suppose $H(L_P) \leq H < \infty$, and $q \in \mathbb{R}^n$ satisfies $\sum_i q_i = d$ and $\beta H \leq \frac{1}{2}$, where $\beta = \|q - p\|_\infty$. Then there is a Parseval W with $\text{diag}(WW^T) = q$ and*

$$\|U - W\|_{\mathbb{F}}^2 \leq \|UU^T - WW^T\|_{\mathbb{F}}^2 \leq \frac{15}{4} \|q - p\|_1.$$

W is obtained from U by a row scaling with ratio of scales at most 2, whitening, and an orthogonal right factor.

Proof. Let s minimise $\Phi(s) = F(s) - \langle q, s \rangle$ over the cube $[0, 1]^n$ (compactness), and write $r = r(s)$. Since $\partial_i \Phi = r_i - q_i$ ([Lemma 2.4](#)), coordinatewise optimality gives

$$r_i \leq q_i \text{ if } s_i > 0, \quad r_i \geq q_i \text{ if } s_i < 1. \tag{2.1}$$

Let $z = e^{2s}$ with median m , so $1 \leq \min z \leq m \leq \max z \leq e^2$. If $z_i > m$ then $s_i > 0$, so by [Lemma 2.5](#) and (2.1) $(Lz)_i \leq z_i(r_i - p_i) \leq z_i(q_i - p_i) \leq \beta z_i$; if $z_i < m$ then $s_i < 1$ and likewise $(Lz^{-1})_i \leq \beta z_i^{-1}$. By [Lemma 2.7](#), $\max z / \min z \leq (1 - \beta H)^{-2} \leq 4$, i.e. $\text{osc}(s) \leq \log 2 < 1$. Hence s does not meet both faces of the cube: either all $s_i > 0$, and then $r \leq q$, or some $s_j = 0$, all $s_i < 1$, and then $r \geq q$. Since $\sum r_i = d = \sum q_i$, in both cases $r = q$.

For the distance let $w = e^s$, $f = q - p$, and let c be the midpoint of the range of z^2 . [Lemma 2.5](#) with $r = q$ gives $Lz \leq z \circ f$, so, using $\sum_i f_i = 0$,

$$z^T Lz \leq \sum_i z_i^2 f_i = \sum_i (z_i^2 - c) f_i \leq \frac{1}{2} \text{osc}(z^2) \|f\|_1.$$

As $(z_i - z_j)^2 \geq 4 \min w^2 (w_i - w_j)^2$, [Lemmas 2.2](#) and [2.3](#) give

$$\|P(s) - P\|_F^2 \leq \frac{2w^T Lw}{\min w^2} \leq \frac{z^T Lz}{2 \min z^2} \leq \frac{\text{osc}(z^2)}{4 \min z^2} \|f\|_1 \leq \frac{4^2 - 1}{4} \|f\|_1.$$

Finally apply [Lemma 2.1](#) to the whitened frame $DU(U^T D^2 U)^{-1/2}$, $D = \text{Diag}(w)$, whose projection is $P(s)$. $\square$

Remark 2.9. The hypothesis forces $q_i \geq p_i/2 > 0$: for $n \geq 2$, the barrier for $S = [n] \setminus \{i\}$ gives $1 \leq (L\psi)_i \leq p_i(1 - p_i)H$, so $\beta \leq 1/(2H) \leq p_i/2$. No existence theorem for scalings, no flow, no genericity and no connectivity argument is used: the box supplies a minimiser, the median barrier keeps it away from one face, and the trace identity $\sum_i r_i = d$ does the rest. With $\beta H < 1$ and a box of side $T > -\log(1 - \beta H)$ the same proof gives scale ratio $(1 - \beta H)^{-1}$ and cost $\frac{1}{4}((1 - \beta H)^{-4} - 1)\|q - p\|_1$.

2.4 Barriers from a dense core

Lemma 2.10 (Dense core with exceptional vertices). *Let L be the Laplacian of nonnegative symmetric weights w_{ij} on $[n]$, and let $[n] = \mathcal{G} \sqcup \mathcal{B}$. Suppose:*

- (a) *there is $\kappa > 0$ with $\sum_{j \in S} w_{ij} \geq \kappa$ for every $i \in \mathcal{G}$ and every $S \subseteq [n] \setminus \{i\}$ with $|S| \geq n/2$;*
- (b) *there are $y \in \mathbb{R}^{\mathcal{B}}$, $y \geq 0$, and $R, F_{\mathcal{B}} \geq 0$ with $\|y\|_{\infty} \leq R$, $(L_{\mathcal{B}\mathcal{B}}y)_i \geq 1$ for $i \in \mathcal{B}$, and $\sum_{j \in \mathcal{B}} w_{ij}y_j \leq F_{\mathcal{B}}$ for $i \in \mathcal{G}$. (If $\mathcal{B} = \emptyset$ take $R = F_{\mathcal{B}} = 0$.)*

Then $H(L) \leq (1 + F_{\mathcal{B}})/\kappa + R$. Moreover, with $b = |\mathcal{B}|$:

- (i) (a) *holds with $\kappa = \vartheta\gamma$ if $\vartheta \in (0, \frac{1}{2}]$, $\gamma > 0$, and every $i \in \mathcal{G}$ has at least $(\frac{1}{2} + \vartheta)n$ indices $j \neq i$ with $w_{ij} \geq \gamma/n$;*
- (ii) (b) *holds with $R = \sqrt{b}/\lambda$, $F_{\mathcal{B}} = b$, if $x^T Lx \geq \lambda\|x\|^2$ for every x supported on $\mathcal{B}$ ($\lambda > 0$);*
- (iii) *for $w_{ij} = P_{ij}^2$, (b) holds with $R = \frac{8}{3\alpha}$, $F_{\mathcal{B}} = \frac{2}{3\alpha} \max_i p_i$ if $\alpha > 0$, $p_i \geq \alpha/2$ on $\mathcal{B}$ and $\sum_{i \in \mathcal{B}} p_i \leq \frac{1}{4}$.*

Proof. Let $|S| \geq n/2$, put $\lambda_0 = (1 + F_{\mathcal{B}})/\kappa$, extend y by 0 to $\mathcal{G}$, and let $\psi = (\lambda_0 \mathbf{1} + y) \circ \mathbf{1}_{S^c}$, so $0 \leq \psi \leq \lambda_0 + R$ and $\psi_j \leq \lambda_0 + y_j$ for all j . For $i \in \mathcal{G} \setminus S$, $\psi_i = \lambda_0$ and $\psi_j = 0$ on S , so

$$(L\psi)_i \geq \lambda_0 \sum_{j \in S} w_{ij} - \sum_{j \in \mathcal{B}} w_{ij}y_j \geq \lambda_0 \kappa - F_{\mathcal{B}} = 1.$$

For $i \in \mathcal{B} \setminus S$, $\psi_i = \lambda_0 + y_i$, so $(L\psi)_i \geq \sum_j w_{ij}(y_i - y_j) = (L_{\mathcal{B}\mathcal{B}}y)_i \geq 1$. Thus ψ is an S -barrier.

(i) If N_i is the set of such j and $S \subseteq [n] \setminus \{i\}$, then $|N_i \cap S| \geq |N_i| + |S| - (n - 1) > \vartheta n$. (ii) $C = L_{\mathcal{B}\mathcal{B}} \succeq \lambda I$; let $y = C^{-1} \mathbf{1}_{\mathcal{B}}$. If y had a negative minimum at i , then $(Cy)_i = y_i \sum_{j \notin \mathcal{B}} w_{ij} + \sum_{j \in \mathcal{B}} w_{ij}(y_i - y_j) \leq 0$, a contradiction; so $y \geq 0$, and $\|y\|_{\infty} \leq \|y\|_2 \leq \sqrt{b}/\lambda$. Since $(C\mathbf{1}_{\mathcal{B}})_j = \sum_{i \in \mathcal{G}} w_{ij}$ for $j \in \mathcal{B}$, for each $i_0 \in \mathcal{G}$

$$\sum_{j \in \mathcal{B}} w_{i_0 j} y_j \leq \sum_{i \in \mathcal{G}} \sum_{j \in \mathcal{B}} w_{ij} y_j = \mathbf{1}_{\mathcal{B}}^T C y = b.$$

(iii) $P_{ij}^2 \leq p_i p_j$ gives, for $i \in \mathcal{B}$, $(L_{\mathcal{B}\mathcal{B}} \mathbf{1}_{\mathcal{B}})_i = p_i - \sum_{j \in \mathcal{B}} P_{ij}^2 \geq p_i(1 - \sum_{j \in \mathcal{B}} p_j) \geq \frac{3\alpha}{8}$, and for $i \in \mathcal{G}$, $\sum_{j \in \mathcal{B}} P_{ij}^2 \leq p_i/4$. Take $y = \frac{8}{3\alpha} \mathbf{1}_{\mathcal{B}}$. $\square$

Remark 2.11 (Poisson constants). The ℓ_∞ Poisson constant K of L is the least K such that every $f \perp \mathbf{1}$ has a solution of $Lg = f$ with $\|g\|_\infty \leq K\|f\|_\infty$. Then $K \leq H \leq 2K$. Indeed, for $|S| \geq n/2$ the forcing $f = \mathbf{1} - \frac{n}{|S|}\mathbf{1}_S$ has $\|f\|_\infty \leq 1$, and $g + K\mathbf{1}$ is an S -barrier; conversely, if $H < \infty$ the graph is connected, so $Lg = f$ is solvable for every $f \perp \mathbf{1}$, and if $\|f\|_\infty \leq 1$ and m is a median of g , the argument of [Lemma 2.7](#) applied to $g - m\mathbf{1} - \beta'\psi$ ($\beta' > 1$) gives $g \leq m + H$, and symmetrically $g \geq m - H$. Barriers are more convenient than Poisson solutions because they can be written down explicitly, as in [Lemma 2.10](#).

2.5 From a seed to an exact frame

Both constructions produce the following object.

Definition 2.12 (Seed). Let $\Theta \geq 2$ and $t > 0$. A (Θ, t) -seed for $X \in \mathbb{R}^{n \times d}$ is a Parseval $V \in \mathbb{R}^{n \times d}$ with

$$\|V - X\|_F^2 \leq \Theta t^2 d, \quad H(L_{VV^T}) \leq \frac{\Theta}{at^2}, \quad \max_i |(VV^T)_{ii} - a| \leq \frac{at^2}{2\Theta}.$$

Corollary 2.13. *If X has a (Θ, t) -seed, there is an ENP frame W with $\|X - W\|_F^2 \leq 3\Theta t^2 d$.*

Proof. Apply [Theorem 2.8](#) to V with $q = a\mathbf{1}$: $\beta H \leq \frac{1}{2}$, and $\|V - W\|_F^2 \leq \frac{15}{4}n \cdot \frac{at^2}{2\Theta} \leq t^2 d$. Then $\|X - W\|_F^2 \leq 2\|X - V\|_F^2 + 2\|V - W\|_F^2 \leq (2\Theta + 2)t^2 d \leq 3\Theta t^2 d$. $\square$

3 Bounded rank: the Hamilton–Moitra estimate

This section reproduces the radial-isotropic and ordered-coordinate argument of Hamilton and Moitra [\[2\]](#), with a direct compactness proof of the existence of the radial scaling. It is used for d below a large absolute constant, and it also shows that ENP frames exist.

Lemma 3.1. *Let $1 \leq d \leq n$, let $X \in \mathbb{R}^{n \times d}$ have $\|x_i\|^2 = a$ for all i , and let $\eta = \|X^T X - I\|_{\text{op}}$. There is an ENP frame W with $\|X - W\|_F^2 \leq \eta d(d-1)$.*

Proof. If $d = 1$ then $X^T X = na = 1$ and $W = X$ works. Let $d \geq 2$.

Step 1: full spark. Suppose first that every $d \times d$ minor $\mu_S = \det(X_S)^2$ ($S \subset [n]$, $|S| = d$) is nonzero. Consider $F(s) = \frac{1}{2} \log \det(X^T \text{Diag}(e^{2s})X) - a \sum_i s_i$. By Cauchy–Binet,

$$F(s) = \frac{1}{2} \log \sum_{|S|=d} \mu_S e^{T(s)_S}, \quad T(s)_S = 2 \left(\sum_{i \in S} s_i - a \sum_i s_i \right),$$

where $T : \mathbb{R}^n \rightarrow \mathbb{R}^{\binom{n}{d}}$ is linear and $\sum_S T(s)_S = 0$ because each i lies in a fraction $d/n = a$ of the d -sets. The function $\mathcal{E}(y) = \sum_S \mu_S e^{y_S}$ restricted to the subspace $\text{im } T$ has the nonempty sublevel set $\{\mathcal{E} \leq \mathcal{E}(0)\}$, on which $y_S \leq \log(\mathcal{E}(0)/\mu_S)$ for every S and hence, by $\sum_S y_S = 0$, also y_S is bounded below. This set is compact, so $\mathcal{E}|_{\text{im } T}$ attains its minimum at some $T(s^*)$, and s^* minimises F .

Let $D = \text{Diag}(e^{s^*})$, $M = (X^T D^2 X)^{-1/2}$ and $W = DXM$. The computation in the proof of [Lemma 2.4](#) (which does not use $X^T X = I$) gives $\partial_i F = \|w_i\|^2 - a$, so W is ENP, and since $w_i = e^{s_i^*} M x_i$ has norm $\sqrt{a}$,

$$w_i = \sqrt{a} \frac{M x_i}{\|M x_i\|}.$$

Step 2: ordered coordinates. Replacing X, W by XO, WO for a suitable $O \in O(d)$ changes neither the hypotheses nor the distance, and makes $M = \text{Diag}(\lambda_1, \dots, \lambda_d)$ with $0 < \lambda_1 \leq \dots \leq \lambda_d$.

Then $w_{ij} = \lambda_j x_{ij} / \sqrt{c_i}$ with $c_i = a^{-1} \sum_\ell \lambda_\ell^2 x_{i\ell}^2$, so w_{ij} has the sign of x_{ij} and $w_{ij}^2 - x_{ij}^2 = x_{ij}^2 (\lambda_j^2 / c_i - 1)$ is nonpositive and then nonnegative as j increases. As $\sum_j (x_{ij}^2 - w_{ij}^2) = a - a = 0$, the prefix sums $\Delta_i(k) = \sum_{j \leq k} (x_{ij}^2 - w_{ij}^2)$ are nonnegative, vanish at $k = d$, and

$$\sum_j |x_{ij}^2 - w_{ij}^2| = 2 \max_k \Delta_i(k) \leq 2 \sum_{k=1}^{d-1} \Delta_i(k).$$

Since $W^T W = I$ and $(X^T X)_{jj} \leq 1 + \eta$, $\sum_i \Delta_i(k) = \sum_{j \leq k} ((X^T X)_{jj} - 1) \leq k\eta$. Equal signs give $(x_{ij} - w_{ij})^2 \leq |x_{ij}^2 - w_{ij}^2|$, and therefore

$$\|X - W\|_F^2 \leq 2\eta \sum_{k=1}^{d-1} k = \eta d(d-1).$$

Step 3: general X . Let G be a Vandermonde matrix with rows $(1, \theta_i, \dots, \theta_i^{d-1})$ for distinct θ_i ; all its maximal minors are nonzero. Each maximal minor of $X + hG$ is a polynomial in h with nonzero leading coefficient, so for all but finitely many h the matrix $X + hG$ is full spark; choose such $h_k \rightarrow 0$ (so small that no row of $X + h_k G$ vanishes) and normalise each row of $X + h_k G$ to length $\sqrt{a}$ (a positive diagonal left factor, which preserves full spark). The resulting $X_k \rightarrow X$ have ENP frames W_k with $\|X_k - W_k\|_F^2 \leq \eta_k d(d-1)$, $\eta_k \rightarrow \eta$. The set of ENP frames is compact; a limit point W of (W_k) is ENP and satisfies the bound. $\square$

Corollary 3.2. *For all $1 \leq d \leq n$ an ENP frame exists. Consequently every equal-norm $X \in \mathbb{R}^{n \times d}$ is within $\|X - W\|_F^2 \leq 2\|X\|_F^2 + 2\|W\|_F^2 = 4d$ of an ENP frame W .*

Proof. Apply Lemma 3.1 to $x_i = \sqrt{a} e_i$ for all i . $\square$

4 The many-row seed

Theorem 4.1. *There are absolute constants $B, c_*, C_* > 0$ such that the following holds. Let $2 \leq d$, $n \geq Bd^2$, let $X \in \mathbb{R}^{n \times d}$ have $\|x_i\|^2 = a = d/n$ for all i , and suppose $\eta := \|X^T X - I\|_{\text{op}}$ satisfies $0 < \eta \leq c_*$. Then there is an ENP frame W with $\|X - W\|_F^2 \leq C_* \eta d$.*

Throughout, $S = X^T X$, so $\|S - I\|_{\text{op}} \leq \eta \leq c_* \leq \frac{1}{2}$ (the choice below gives $c_* \leq \frac{1}{64}$), and $e_i = x_i / \sqrt{a}$.

4.1 The noise

In the Frobenius space $\mathbb{R}^{n \times d}$ consider the subspaces

$$\mathcal{E} = \{Z : \langle x_i, z_i \rangle = 0 \ \forall i\}, \quad \mathcal{F} = \{Z \in \mathcal{E} : X^T Z + Z^T X = 0\},$$

and let $m = \dim \mathcal{E} - \dim \mathcal{F} \leq \dim \text{Sym}_d \leq d^2 \leq n$. For a standard Gaussian $g \in \mathbb{R}^{n \times d}$ put

$$Z_0 = n^{-1/2} \Pi_{\mathcal{E}} g, \quad Z = n^{-1/2} \Pi_{\mathcal{F}} g, \quad Z_\perp = Z_0 - Z = n^{-1/2} \Pi_{\mathcal{E} \ominus \mathcal{F}} g.$$

Z and $Z_\perp$ are independent. The rows $z_{0,i} = n^{-1/2} (I - e_i e_i^T) g_i$ of Z_0 are independent. Since $\text{Cov}(Z) \preceq \text{Cov}(Z_0)$, each row z_i of Z is Gaussian with covariance $\Sigma_i \preceq n^{-1} (I - e_i e_i^T)$, and $\text{Cov}(\text{vec } Z) \preceq I/n$. Moreover

$$\mathbb{E} \|Z\|_F^2 = \frac{n(d-1) - m}{n} \leq d, \quad \mathbb{E} \|Z - Z_0\|_F^2 = \frac{m}{n} \leq \frac{d^2}{n}. \quad (4.1)$$

Lemma 4.2 (Row moments). *Let $r_i = \|z_i\|^2/a$, $\mu_i = \mathbb{E}r_i$ and $\delta_i = 1 - \mu_i$. Then $1/d \leq \delta_i \leq 1$, $\sum_i a\delta_i = 1 + m/n \leq 2$, and*

$$\mathcal{M} := a \sum_i \mathbb{E}[(r_i - 1)^2(1 + r_i)^2] \leq C_{\mathcal{M}},$$

with $C_{\mathcal{M}}$ absolute.

Proof. $\tilde{\Sigma}_i = \Sigma_i/a \preceq d^{-1}I$ has trace $\mu_i \leq (d-1)/d$, so $\text{tr } \tilde{\Sigma}_i^2 \leq 1/d$ and $\text{tr } \tilde{\Sigma}_i^4 \leq d^{-3}$. For $\xi_i = r_i - \mu_i$ the Gaussian moment formulas give $\mathbb{E}\xi_i^2 = 2 \text{tr } \tilde{\Sigma}_i^2 \leq 2/d$ and $\mathbb{E}\xi_i^4 = 12(\text{tr } \tilde{\Sigma}_i^2)^2 + 48 \text{tr } \tilde{\Sigma}_i^4 \leq 60/d^2$. Next $\sum_i a\delta_i = d - \mathbb{E}\|Z\|_{\text{F}}^2 = 1 + m/n$ by (4.1). Finally $(r_i - 1)^2 \leq 2\xi_i^2 + 2\delta_i^2$ and $(1 + r_i)^2 \leq 8 + 2\xi_i^2$, so

$$\mathbb{E}(r_i - 1)^2(1 + r_i)^2 \leq 16\mathbb{E}\xi_i^2 + 4\mathbb{E}\xi_i^4 + 16\delta_i^2 + 4\delta_i^2\mathbb{E}\xi_i^2 \leq \frac{C}{d} + C\delta_i^2,$$

and $a \sum_i (C/d + C\delta_i^2) \leq C + Ca \sum_i \delta_i \leq 3C$. $\square$

4.2 Row normalisation and the exact second-order identity

For $0 < t \leq 1$ set

$$v_i = \frac{x_i + tz_i}{\sqrt{1 + t^2 r_i}}, \quad v_{0,i} = \frac{x_i + tz_{0,i}}{\sqrt{1 + t^2 r_{0,i}}}, \quad r_{0,i} = \|z_{0,i}\|^2/a.$$

Since $z_i \perp x_i$, $\|v_i\|^2 = \|v_{0,i}\|^2 = a$ exactly. The map $y \mapsto \sqrt{a}y/\|y\|$ is the metric projection onto the closed ball of radius $\sqrt{a}$ on $\{\|y\| \geq \sqrt{a}\}$, hence 1-Lipschitz there. Thus

$$\|V - X\|_{\text{F}} \leq t\|Z\|_{\text{F}}, \quad \|V - V_0\|_{\text{F}} \leq t\|Z - Z_0\|_{\text{F}}, \quad \|V\|_{\text{op}} \leq \|X + tZ\|_{\text{op}}, \quad \|V_0\|_{\text{op}} \leq \|X + tZ_0\|_{\text{op}}, \quad (4.2)$$

the last two because $V = \text{Diag}(\lambda)(X + tZ)$ with $0 < \lambda_i \leq 1$, and similarly for V_0 .

Expanding $1/(1 + t^2 r_i) = 1 - t^2 c_i$ with

$$c_i = \frac{r_i}{1 + t^2 r_i},$$

and using $X^T Z + Z^T X = 0$, one finds the exact identity

$$V^T V = S + t^2 Q(Z) + R_3 + R_4, \quad Q(Z) = Z^T Z - \sum_i r_i x_i x_i^T, \quad (4.3)$$

$$R_3 = -t^3 \sum_i c_i (x_i z_i^T + z_i x_i^T), \quad R_4 = -t^4 \sum_i c_i (z_i z_i^T - r_i x_i x_i^T).$$

Lemma 4.3 (Mean and fluctuation of Q). $\|\mathbb{E}Q(Z) - (I - S)\|_{\text{op}} \leq m/n$ and $\mathbb{E}\|Q(Z) - \mathbb{E}Q(Z)\|_{\text{F}}^2 \leq 8d^2/n$.

Proof. For Z_0 : $\mathbb{E}Z_0^T Z_0 = n^{-1} \sum_i (I - e_i e_i^T) = I - S/d$ and $\mathbb{E}r_{0,i} = (d-1)/d$, so $\mathbb{E}Q(Z_0) = I - S/d - (1 - 1/d)S = I - S$. As $Z_0 = Z + Z_{\perp}$ with independent centred summands and Q is quadratic, $\mathbb{E}Q(Z) = \mathbb{E}Q(Z_0) - \mathbb{E}Q(Z_{\perp})$, and $\mathbb{E}Q(Z_{\perp})$ is the difference of the positive semidefinite matrices $\mathbb{E}Z_{\perp}^T Z_{\perp}$ and $\sum_i \mathbb{E}(\|z_{\perp,i}\|^2/a) x_i x_i^T$, both of trace $\mathbb{E}\|Z_{\perp}\|_{\text{F}}^2 = m/n$; so its norm is at most m/n .

Entry (α, β) of Q is the quadratic form $\sum_i z_i^T B_{\alpha\beta,i} z_i$ with $B_{\alpha\beta,i} = \frac{1}{2}(E_{\alpha\beta} + E_{\beta\alpha}) - (e_i)_{\alpha}(e_i)_{\beta}I$. For a Gaussian vector with covariance $\Sigma \preceq I/n$ and symmetric coefficient matrix $\mathbf{B}$, the variance of the quadratic form is $2\|\Sigma^{1/2}\mathbf{B}\Sigma^{1/2}\|_{\text{F}}^2 \leq 2\|\mathbf{B}\|_{\text{F}}^2/n^2$. Since $\sum_{\alpha,\beta} \|B_{\alpha\beta,i}\|_{\text{F}}^2 \leq 2d^2 + 2d \leq 4d^2$ for each i , summing over α, β and the n blocks gives the claim. $\square$

Lemma 4.4 (Remainder). *Let $\bar{c} = 1/(1+t^2)$, $\Gamma = \text{Diag}(c_i - \bar{c})$ and $\Lambda = \text{Diag}(c_i r_i - \bar{c})$. Then*

$$\|R_3\|_{\text{op}} \leq 2t^3 \|X\|_{\text{op}} \|\Gamma Z\|_{\text{F}}, \quad \|R_4\|_{\text{op}} \leq t^4 (\|Z\|_{\text{op}}^2 + \|Z\|_{\text{op}} \|\Gamma Z\|_{\text{F}} + \|S\|_{\text{op}} + \|X\|_{\text{op}} \|\Lambda X\|_{\text{F}}),$$

and $\mathbb{E}\|\Gamma Z\|_{\text{F}}^2 \leq \mathcal{M}$, $\mathbb{E}\|\Lambda X\|_{\text{F}}^2 \leq \mathcal{M}$.

Proof. Because $\sum_i (x_i z_i^T + z_i x_i^T) = X^T Z + Z^T X = 0$, the constant part $\bar{c}$ of the weights drops out of R_3 : $R_3 = -t^3 (X^T \Gamma Z + Z^T \Gamma X)$, which gives the first bound. Next $\sum_i c_i z_i z_i^T = \bar{c} Z^T Z + Z^T \Gamma Z$ and $\sum_i c_i r_i x_i x_i^T = \bar{c} S + X^T \Lambda X$, with $\|Z^T \Gamma Z\|_{\text{op}} \leq \|Z\|_{\text{op}} \|\Gamma Z\|_{\text{F}}$ and similarly for Λ , and $\bar{c} \leq 1$. For the moments,

$$c_i - \bar{c} = \frac{r_i - 1}{(1+t^2 r_i)(1+t^2)}, \quad c_i r_i - \bar{c} = \frac{(r_i - 1)(1+r_i+t^2 r_i)}{(1+t^2 r_i)(1+t^2)},$$

so $|c_i - \bar{c}| \leq |r_i - 1|$ and, as $1+r+t^2 r \leq (1+r)(1+t^2 r)$, $|c_i r_i - \bar{c}| \leq |r_i - 1|(1+r_i)$. Hence $\|\Gamma Z\|_{\text{F}}^2 = a \sum_i (c_i - \bar{c})^2 r_i \leq a \sum_i (r_i - 1)^2 (1+r_i)^2$ and $\|\Lambda X\|_{\text{F}}^2 = a \sum_i (c_i r_i - \bar{c})^2 \leq a \sum_i (r_i - 1)^2 (1+r_i)^2$; take expectations and use Lemma 4.2. $\square$

Remark 4.5. The termwise bound $\|R_3\|_{\text{op}} \leq 2t^3 \sum_i c_i \|x_i\| \|z_i\| \approx 2t^3 d$ would only allow $\eta \lesssim d^{-2}$. Centring the weights at $\bar{c}$, which is allowed by the first-order constraint, replaces the n terms of size a by fluctuations of relative size $d^{-1/2}$, whose ℓ_2 -sum over the rows is $O(1)$.

4.3 A dense reference graph

Lemma 4.6. *There are absolute $h, t_0 \in (0, 1]$ and $c_1 > 0$ such that for $d \geq 2$ and $t \leq t_0$, with probability at least $1 - ne^{-c_1(n-1)}$, every i has at least $0.9(n-1)$ indices $j \neq i$ with*

$$|\langle v_{0,i}, v_{0,j} \rangle| \geq h t \sqrt{a/n}.$$

Proof. Write $z_{0,j}/\sqrt{a} = d^{-1/2} g'_j$ with $g'_j \sim N(0, I - e_j e_j^T)$ independent. Fix i and condition on row i ; let $f = v_{0,i}/\sqrt{a}$, a unit vector. For $j \neq i$,

$$\frac{\langle v_{0,i}, v_{0,j} \rangle}{a} = \frac{\nu_j}{N_j}, \quad \nu_j = u_j + \frac{t}{\sqrt{d}} \langle f, g'_j \rangle, \quad u_j = \langle f, e_j \rangle, \quad N_j = (1 + \frac{t^2}{d} \|g'_j\|^2)^{1/2} \geq 1,$$

and the target inequality reads $|\nu_j|/N_j \geq ht/\sqrt{d}$. By Markov, $\mathbb{P}(N_j > 2) \leq t^2/3$. On $\{N_j \leq 2\}$ failure requires $|\nu_j| < 2ht/\sqrt{d}$. Here $\nu_j \sim N(u_j, t^2(1 - u_j^2)/d)$. If $|u_j| \leq \frac{1}{2}$ the variance is at least $3t^2/(4d)$ and the Gaussian density bound gives probability at most $4h/\sqrt{3\pi/2} \leq 3h$. If $|u_j| > \frac{1}{2}$ and $2ht \leq \frac{1}{4}$, failure requires $\frac{t}{\sqrt{d}} |\langle f, g'_j \rangle| > \frac{1}{4}$, of probability at most $16t^2/d$. Choose h and t_0 so that $3h + t_0^2/3 + 8t_0^2 \leq \frac{1}{50}$ (this also gives $2ht \leq \frac{1}{4}$, and $16t^2/d \leq 8t^2$ as $d \geq 2$). The failure indicators for $j \neq i$ are conditionally independent with probabilities at most $\frac{1}{50}$, so by the exponential Markov inequality (with parameter 1) more than $0.1(n-1)$ failures have probability at most $\exp(-(n-1)[0.1 - \frac{e-1}{50}]) \leq e^{-c_1(n-1)}$. Take a union bound over i . $\square$

4.4 Proof of Theorem 4.1

Constants. Fix h, t_0, c_1 from Lemma 4.6. Put $\kappa_0 = h^2/200$, $C' = 36^2 \cdot 20$, and $C_{\text{mix}} = 125/h^2 + 3$, then $\Theta = \max\{42, C_{\text{mix}}\}$ and $\theta = \min\{1/(4\Theta), h/60\}$. Let $C_R = 300 + 100\sqrt{C_{\mathcal{M}}}$, choose $t_1 = \min\{t_0, \theta/(16C_R)\}$, and set

$$t^2 = 4\eta/\theta, \quad c_* = \theta t_1^2/4,$$

so that $\eta \leq c_*$ gives $t \leq t_1$. Finally choose B so large that $n \geq Bd^2$ implies $d^2/n \leq \theta/4$, $\sqrt{160d^2/n} \leq \theta/4$, $2aC'd/\kappa_0 \leq \frac{1}{4}$ (this bounds $\sum_{i \in B_0} 2a$ for the set B_0 below, which has $|B_0| \leq C'd/\kappa_0$), $n \geq 30$, and $ne^{-c_1(n-1)} \leq 0.01$.

The event. By Markov's inequality, (4.1), Lemmas 4.3 and 4.4, the net bound (Lemma A.1(c)) for Z and Z_0 , and Lemma 4.6, with positive probability all of the following hold:

$$\|Z\|_F^2 \leq 20d, \quad \|Z - Z_0\|_F^2 \leq \frac{20d^2}{n}, \quad \|Q - \mathbb{E}Q\|_F^2 \leq \frac{160d^2}{n}, \quad \|\Gamma Z\|_F^2 \leq 20\mathcal{M}, \quad \|\Lambda X\|_F^2 \leq 20\mathcal{M},$$

$\|Z\|_{\text{op}}, \|Z_0\|_{\text{op}} \leq 16$, and the conclusion of Lemma 4.6: the five Markov events fail with probability at most $\frac{1}{20}$ each, the two net bounds (Lemma A.1(c)) with $\sigma^2 = 1/n$, $u = 16$) with probability at most $2 \cdot 5^{n+d}e^{-8n}$ each, and Lemma 4.6 with probability at most 0.01. Fix such a sample.

Spectral error. By (4.3), $V^TV - I = (1-t^2)(S-I) + t^2(\mathbb{E}Q - (I-S)) + t^2(Q - \mathbb{E}Q) + R_3 + R_4$. Since $\|X\|_{\text{op}} \leq \sqrt{2}$, $\|S\|_{\text{op}} \leq \frac{3}{2}$ and $t \leq 1$, Lemma 4.4 and the event give $\|R_3\|_{\text{op}} + \|R_4\|_{\text{op}} \leq t^3(2\sqrt{40\mathcal{M}} + 256 + 16\sqrt{20\mathcal{M}} + \frac{3}{2} + \sqrt{40\mathcal{M}}) \leq C_R t^3$, and therefore

$$\delta := \|V^TV - I\|_{\text{op}} \leq \eta + t^2 \frac{d^2}{n} + t^2 \sqrt{\frac{160d^2}{n}} + C_R t^3 \leq \frac{\theta}{4} t^2 + \frac{\theta}{4} t^2 + \frac{\theta}{4} t^2 + \frac{\theta}{16} t^2 < \theta t^2.$$

Graph. By (4.2), $\|V\|_{\text{op}}, \|V_0\|_{\text{op}} \leq \sqrt{2} + 16 \leq 18$ and $\|VV^T - V_0V_0^T\|_F \leq 36t\|Z - Z_0\|_F$, so $\|VV^T - V_0V_0^T\|_F^2 \leq C't^2d^2/n$. Let B_0 be the set of rows whose contribution to this sum exceeds $\kappa_0 at^2$; then $|B_0| \leq C'd/\kappa_0$. A row $i \notin B_0$ has at most $\kappa_0 at^2 / ((h/2)^2 t^2 a/n) = n/50$ indices with $|(VV^T - V_0V_0^T)_{ij}| \geq \frac{h}{2} t \sqrt{a/n}$, so by Lemma 4.6 it has at least $0.85n$ indices $j \neq i$ with $|(VV^T)_{ij}| \geq \frac{h}{2} t \sqrt{a/n}$.

Let $U = V(V^TV)^{-1/2}$, $P = UU^T$. By Lemma 2.1 (rows of $P - VV^T$) a row loses at most $6a\theta^2 t^4 / ((h/4)^2 t^2 a/n) \leq n/20$ of these indices, so each $i \notin B_0$ has at least $0.8n$ indices $j \neq i$ with $P_{ij}^2 \geq (h^2/16)at^2/n$. Also $a/2 \leq P_{ii} \leq 2a$ and $\sum_{i \in B_0} P_{ii} \leq \frac{1}{4}$. Lemma 2.10 with $\mathcal{B} = B_0$, part (i) with $\vartheta = 0.3$ and $\gamma = h^2 at^2/16$, and part (iii) with $\alpha = a$ (so $R = \frac{8}{3a}$, $F_{\mathcal{B}} \leq \frac{4}{3}$) shows that $H(L_P) \leq \frac{7/3}{0.3\gamma} + \frac{8}{3a} \leq C_{\text{mix}}/(at^2)$.

Seed. $\|U - X\|_F^2 \leq 2d\delta^2 + 2t^2\|Z\|_F^2 \leq 42t^2d$, the diagonal error is at most $2a\delta \leq 2\theta at^2 \leq at^2/(2\Theta)$, and the barrier constant is at most $\Theta/(at^2)$. So U is a (Θ, t) -seed, and Corollary 2.13 gives an ENP frame within $3\Theta t^2 d = (12\Theta/\theta)\eta d$ of X . $\square$

5 The moderate-row seed

Theorem 5.1. *There are absolute constants $A, C_m, \varepsilon_0 > 0$ such that the following holds. Let $U \in \mathbb{R}^{n \times d}$ be Parseval with $2d \leq n$, $d \geq A \log(2n)$, and $|||u_i||^2 - a| \leq \varepsilon a$ for all i , where $0 < \varepsilon \leq \varepsilon_0$. Then there is an ENP frame $\widehat{U}$ with $\|\widehat{U} - U\|_F^2 \leq C_m \varepsilon d$.*

Throughout this section $P = UU^T$, $p = \text{diag } P$, $D_p = \text{Diag}(p)$, $L = L_P$, and $\varepsilon_0 \leq \frac{1}{2}$, so $\frac{a}{2} \leq p_i \leq \frac{3a}{2} \leq \frac{3}{4}$. For a symmetric $M \in \mathbb{R}^{n \times n}$, $\mathcal{L}(M) = \sum_{i < j} M_{ij}^2 (e_i - e_j)(e_i - e_j)^T$ is the Laplacian with weights M_{ij}^2 ; thus $L = \mathcal{L}(P)$, $x^T \mathcal{L}(M) x = \frac{1}{2} \|\text{Diag } x, M\|_F^2$, and

$$\mathcal{L}(M) \preceq 2 \max_i \|M_i\|^2 I, \quad \mathcal{L}(M + M') \succeq \frac{1}{2} \mathcal{L}(M) - \mathcal{L}(M'), \quad (5.1)$$

by $(x_i - x_j)^2 \leq 2x_i^2 + 2x_j^2$ and $(v + w)^2 \geq \frac{1}{2}v^2 - w^2$. We write $K = nI - \mathbf{1}\mathbf{1}^T$ for the Laplacian of the complete graph.

Scales. Put $\sigma = at^2$, where $t \asymp \sqrt{\varepsilon}$ is the size of the perturbation. The perturbation creates edges of weight $\gtrsim \sigma/n$, the resulting barrier constant is $O(1/\sigma)$, and so all diagonal errors must be at most a small multiple of σ . The table lists the errors and how each is controlled ($\rho = D^2 t^2$ is the filter level, D a large constant).

diagonal error	size/ σ	reason
input	ε/t^2	$t^2 = K_0 \varepsilon$, K_0 large
linear term $2tAZ$	$6D\sqrt{\log(2n)/d}$	filter (Lemma 5.3), $d \geq A \log 2n$
quadratic fluctuation	η'	concentration, $d \geq A \log 2n$
filter bias $t^2 b_*$	0	cancelled by the drift (Lemmas 5.6 and 5.10)
drift cross terms	$40/D + 1/(2D^2)$	D large
base bias $t^2 b_0$	$12\varepsilon/d$	Lemma 5.6
retraction remainder	$C_E t$	t small

5.1 The horizontal tangent space and the filtered covariance

Fix an isometry $V \in \mathbb{R}^{n \times (n-d)}$ onto $(\text{im } U)^\perp$, and let $\mathcal{Z} = \mathbb{R}^{(n-d) \times d}$ with the Frobenius inner product. For $Z \in \mathcal{Z}$ the matrix VZ is horizontal ($U^T VZ = 0$), and we put

$$Y_Z = VZU^T + UZ^T V^T, \quad \|(Y_Z)_i\|^2 = \|Z^T v_i\|^2 + \|Zu_i\|^2 \leq \|Z\|_{\text{op}}^2, \quad \|Y_Z\|_{\text{F}}^2 = 2\|Z\|_{\text{F}}^2.$$

Define

$$\mathcal{A} : \mathcal{Z} \rightarrow \mathbb{R}^n, \quad (\mathcal{A}Z)_i = v_i^T Z u_i = (VZU^T)_{ii}, \quad \mathcal{A}^* x = N_x := V^T \text{Diag}(x)U.$$

Then $\mathcal{A}\mathcal{A}^* = L$, i.e. $\|N_x\|_{\text{F}}^2 = x^T Lx$ (Lemma 2.2). Let

$$\bar{\mathcal{A}} = D_p^{-1/2} \mathcal{A}, \quad S = \bar{\mathcal{A}} \bar{\mathcal{A}}^* = D_p^{-1/2} L D_p^{-1/2}, \quad \Omega = \bar{\mathcal{A}}^* \bar{\mathcal{A}}.$$

Since $L \succeq 0$ and $S = I - D_p^{-1/2}(P \circ P)D_p^{-1/2}$ with $P \circ P \succeq 0$ (Schur product theorem),

$$0 \preceq S \preceq I, \quad 0 \preceq \Omega \preceq I, \quad \text{tr}(I - S) = \sum_i \frac{P_{ii}^2}{p_i} = d. \quad (5.2)$$

Fix an orthonormal eigenbasis of S , and for each of its members y with eigenvalue $\mu > 0$ put $f_y = D_p^{-1/2} y$ and $Z_y = \mu^{-1/2} \bar{\mathcal{A}}^* y = \mu^{-1/2} N_{f_y}$. Since $BZ_y = \mu Z_y$ and $\langle Z_y, Z_{y'} \rangle = \langle y, Sy' \rangle / \sqrt{\mu \mu'}$, the Z_y form an orthonormal eigenbasis of Ω on $(\ker \Omega)^\perp$. Sums $\sum_{\mu > 0}$ below run over these basis members (with multiplicity). Since $p_i \geq a/2$,

$$\|f_y\|_\infty \leq \sqrt{2/a}. \quad (5.3)$$

Definition 5.2 (Filtered covariance). For $\rho > 0$ let $C_\rho = \rho(I - \Omega)(\rho I + \Omega)^{-1}$ on $\mathcal{Z}$, let $Z \sim N(0, C_\rho/n)$, and $Y = Y_Z$.

Lemma 5.3. (a) $0 \preceq C_\rho \preceq I$, and $C_\rho = I$ on $\ker \Omega$.

(b) $\text{Cov}(\mathcal{A}Z) = n^{-1} D_p^{1/2} \rho S (I - S) (\rho I + S)^{-1} D_p^{1/2} \preceq (\rho/n) D_p$.

(c) $I - C_\rho = \Omega + R_\rho$ with $R_\rho = \sum_{\mu > 0} r_\mu Z_y \otimes Z_y$, $r_\mu = \mu(1 - \mu)/(\rho + \mu) \geq 0$, and

$$\sum_{\mu > 0} r_\mu \leq d, \quad \sum_{\mu > 0} \frac{r_\mu}{\sqrt{\mu}} \leq \frac{d}{2\sqrt{\rho}}, \quad \sum_{\mu > 0} \frac{r_\mu}{\mu} \leq \frac{d}{\rho}.$$

Proof. (a) The eigenvalues of C_ρ are $\rho(1 - \mu)/(\rho + \mu) \in [0, 1]$. (b) By singular value calculus $\tilde{\mathcal{A}}C_\rho\tilde{\mathcal{A}}^* = \rho S(I - S)(\rho I + S)^{-1}$, whose eigenvalues $\rho\mu(1 - \mu)/(\rho + \mu)$ are at most ρ . (c) On the μ -eigenspace, $1 - \rho(1 - \mu)/(\rho + \mu) = \mu + \mu(1 - \mu)/(\rho + \mu)$. The sums follow from $\mu/(\rho + \mu) \leq 1$, $\sqrt{\mu}/(\rho + \mu) \leq 1/(2\sqrt{\rho})$, $1/(\rho + \mu) \leq 1/\rho$, and $\sum_{\mu>0}(1 - \mu) \leq \text{tr}(I - S) = d$. $\square$

Remark 5.4. Only these properties of the filter $\phi(\mu) = \rho(1 - \mu)/(\rho + \mu)$ are used: $\phi(0) = 1$ (so $C_\rho = I$ on $\ker \Omega$), $0 \leq \phi(\mu) \leq 1 - \mu$, $\mu\phi(\mu) \leq \rho$, and the sums in (c). The hard cutoff $(1 - \mu)\mathbf{1}[\mu < \rho]$ satisfies them with $\sum r_\mu/\sqrt{\mu} \leq d/\sqrt{\rho}$, which doubles some constants below; the rational filter keeps every object a rational function of S , which is convenient for formalisation.

5.2 Commutator identities

For $Z \in \mathcal{Z}$ define the quadratic diagonal and its polarisation

$$q_i(Z) = \|Z^T v_i\|^2 - \|Z u_i\|^2, \quad q_i(Z, M) = \langle Z^T v_i, M^T v_i \rangle - \langle Z u_i, M u_i \rangle,$$

so that $q(Z + M) = q(Z) + 2q(Z, M) + q(M)$ and, by Cauchy–Schwarz,

$$|q_i(Z, M)| \leq \|(Y_Z)_i\| \|(Y_M)_i\|, \quad |q_i(M)| \leq \|(Y_M)_i\|^2. \quad (5.4)$$

Lemma 5.5. *Let $f \in \mathbb{R}^n$, $F = \text{Diag}(f)$, and $R = I - 2P$ (an orthogonal matrix). Then:*

- (a) $q(N_f) = \mathcal{A}\Gamma_f$ with $\Gamma_f = V^T F R F U$, and $\|\Gamma_f\|_F \leq 2\|f\|_\infty \|N_f\|_F$;
- (b) $T_f := Y_{N_f} = F P + P F - 2 P F P$ satisfies $[X, T_f] = R F [X, P] + [X, P] F R$ for diagonal X ; hence $\mathcal{L}(T_f) \preceq 4\|f\|_\infty^2 L$, and $x^T \mathcal{L}(T_{e_k}) x \leq 2 \sum_j P_{kj}^2 (x_k - x_j)^2$.

Proof. (a) For $x \in \mathbb{R}^n$ put $X = \text{Diag}(x)$, $D_x = V^T X V$, $A_x = U^T X U$. Then $\langle x, q(Z) \rangle = \langle Z, D_x Z - Z A_x \rangle$ for every Z , and, since X and F commute,

$$D_x N_f - N_f A_x = V^T [X(I - P)F - F P X] U = V^T [F(I - P)X - X P F] U = D_f N_x - N_x A_f.$$

Hence $\langle x, q(N_f) \rangle = \langle N_f, D_f N_x - N_x A_f \rangle = \langle D_f N_f - N_f A_f, N_x \rangle$. As $\langle M, N_x \rangle = \sum_i x_i v_i^T M u_i = \langle \mathcal{A}M, x \rangle$ for every M , this says $q(N_f) = \mathcal{A}(D_f N_f - N_f A_f)$, and $D_f N_f - N_f A_f = V^T F(I - P) F U - V^T F P F U = \Gamma_f$. The norm bound follows from $\|D_f\|_{\text{op}}, \|A_f\|_{\text{op}} \leq \|f\|_\infty$.

(b) $Y_{N_f} = V V^T F U U^T + U U^T F V V^T = (I - P) F P + P F (I - P)$. Since X commutes with F , $[X, F P] = F [X, P]$, $[X, P F] = [X, P] F$ and $[X, P F P] = [X, P] F P + P F [X, P]$, which gives the commutator formula. Then $\|[X, T_f]\|_F \leq 2\|f\|_\infty \|[X, P]\|_F$, i.e. the first bound (Lemma 2.2). For $f = e_k$, $E_k[X, P]$ and $[X, P]E_k$ are the k th row and column of $[X, P]$, each of squared norm $\sum_j P_{kj}^2 (x_k - x_j)^2$. $\square$

5.3 The second-order mean and the drift

Lemma 5.6 (Mean). $\mathbb{E}q(Z) = a\mathbf{1} - p - b_0 - b_*$ with $b_* := \mathcal{A}m_*$, where

$$b_0 = \frac{1}{n} L(p^{-1}), \quad \|b_0\|_\infty \leq \frac{12\varepsilon}{n}, \quad m_* = \frac{1}{n} \sum_{\mu>0} \frac{r_\mu}{\mu} \Gamma_{f_y}, \quad \|m_*\|_F^2 \leq \frac{2a}{\rho}.$$

Proof. For $G \sim N(0, I)$ on $\mathcal{Z}$, $\mathbb{E}\|G^T v_i\|^2 = d(1 - p_i)$ and $\mathbb{E}\|Gu_i\|^2 = (n - d)p_i$, so the unfiltered covariance I/n contributes $a\mathbf{1} - p$. The covariance $\Omega = \sum_j \zeta_j \otimes \zeta_j$, $\zeta_j = \bar{\mathcal{A}}^* e_j = v_j u_j^T / \sqrt{p_j}$, contributes

$$\frac{1}{n} \sum_j q_i(\zeta_j) = \frac{1}{n} \sum_j \frac{(I - P)_{ij}^2 p_j - (1 - p_j) P_{ij}^2}{p_j} = \frac{1}{n} \left(1 - \sum_j \frac{P_{ij}^2}{p_j} \right) = \frac{1}{n} (Lp^{-1})_i,$$

using $\sum_j (I - P)_{ij}^2 = 1 - p_i$ and $\sum_j P_{ij}^2 = p_i$. Here $(Lp^{-1})_i = \sum_j P_{ij}^2 (p_i^{-1} - p_j^{-1})$ and $|p_i^{-1} - p_j^{-1}| \leq 2\varepsilon a / ((1 - \varepsilon)a)^2 \leq 8\varepsilon/a$, so $|(Lp^{-1})_i| \leq 12\varepsilon$. Finally R_ρ/n contributes $\frac{1}{n} \sum_\mu r_\mu q(Z_y) = \frac{1}{n} \sum_\mu \frac{r_\mu}{\mu} q(N_{f_y}) = \mathcal{A}m_*$ by Lemma 5.5(a), and with $\|N_{f_y}\|_F = \sqrt{\mu}$, (5.3) and Lemma 5.3(c),

$$\|m_*\|_F \leq \frac{2}{n} \sqrt{\frac{2}{a}} \sum_{\mu > 0} \frac{r_\mu}{\sqrt{\mu}} \leq \frac{2}{n} \sqrt{\frac{2}{a}} \cdot \frac{d}{2\sqrt{\rho}} = \sqrt{\frac{2a}{\rho}}. \quad \square$$

Remark 5.7 (Why a drift). The filter removes covariance from the directions Z_y , and with it their share $t^2 b_* = t^2 \mathcal{A}m_*$ of the second-order drift towards $a\mathbf{1}$. This bias need not be below the tolerance $\sigma/(12\Theta)$ used later: for instance, for many small blocks joined through a common row it is of order at^2 at that row. (Since m_* does not depend on the choice of eigenbasis, it can also be written $m_* = \frac{1}{n} V^T [(D_p^{-1/2} (I - S)(\rho I + S)^{-1} D_p^{-1/2}) \circ (I - 2P)] U$.) But it is the *first-order* diagonal change of the deterministic horizontal direction $\frac{t}{2} m_*$, of norm $\frac{t}{2} \|m_*\|_F \leq \sqrt{a/2}/D$. Adding this direction to the noise cancels the bias exactly, at the price of cross terms of size $O(\sigma/D)$.

5.4 The expected graph

Since $\mathbb{E}Y_{ij}^2$ is linear in the covariance, $\mathbb{E}_\Sigma \mathcal{L}(Y) = \sum_k \lambda_k \mathcal{L}(Y_{\psi_k})$ for every representation $\Sigma = \sum_k \lambda_k \psi_k \otimes \psi_k$ with real λ_k ; we use it for $C_\rho/n = (I - \Omega - R_\rho)/n$ with $\Omega = \sum_j \zeta_j \otimes \zeta_j$.

Lemma 5.8. $\mathbb{E}\mathcal{L}(Y) \succeq \frac{a}{4n} \mathbf{K} - \left(\frac{2}{n} + \frac{8}{d} + \frac{8}{\rho} \right) L$.

Proof. Unfiltered noise. For $Z_0 \sim N(0, I/n)$ and $i \neq j$, $(Y_{Z_0})_{ij} = v_i^T Z_0 u_j + v_j^T Z_0 u_i$ has variance $n^{-1} \|v_i u_j^T + v_j u_i^T\|_F^2 = n^{-1} (p_i + p_j - 2p_i p_j - 2P_{ij}^2)$. As $p \leq \frac{3}{4}$, $p_i + p_j - 2p_i p_j \geq \frac{1}{4} (p_i + p_j) \geq \frac{a}{4}$, so $\mathbb{E}\mathcal{L}(Y_{Z_0}) \succeq \frac{a}{4n} \mathbf{K} - \frac{2}{n} L$.

Base loss. $Y_{\zeta_k} = T_{e_k} / \sqrt{p_k}$, so by Lemma 5.5(b) and $p_k \geq a/2$, $\sum_k x^T \mathcal{L}(Y_{\zeta_k}) x \leq \frac{2}{a} \cdot 2 \sum_{k,j} P_{kj}^2 (x_k - x_j)^2 = \frac{8}{a} x^T L x$; the base part Ω/n therefore removes at most $\frac{8}{d} L$.

Residual loss. $\mathcal{L}(Y_{Z_y}) = \mu^{-1} \mathcal{L}(T_{f_y}) \preceq \frac{8}{a\mu} L$ by Lemma 5.5(b) and (5.3), so R_ρ/n removes at most $\frac{8}{an} \sum_\mu \frac{r_\mu}{\mu} L \leq \frac{8}{\rho} L$. $\square$

5.5 A good sample

Let $\ell_i = \frac{1}{n} \sum_\mu r_\mu \|(Y_{Z_y})_i\|^2$ be the variance that R_ρ/n removes from row i . Since $\|Y_{Z_y}\|_F^2 = 2$, $\sum_i \ell_i \leq \frac{2}{n} \sum_\mu r_\mu \leq 2a$. Fix $\delta_0 = 1/6400$ and let

$$\mathfrak{B} = \{i : \ell_i > \delta_0 a\}, \quad b = |\mathfrak{B}| \leq 2/\delta_0 =: b_{\max},$$

a deterministic set depending only on U and ρ . Write $\mathcal{C}(Y) = \sum_{i < j} 2P_{ij} Y_{ij} (e_i - e_j)(e_i - e_j)^T$, so that $\mathcal{L}(P + tY) = L + t\mathcal{C}(Y) + t^2 \mathcal{L}(Y)$.

Lemma 5.9. *There is an absolute $h \in (0, 1)$ and, for each $\eta' \in (0, \frac{1}{32}]$, a constant $A_{\eta'}$ such that the following holds for all $\rho > 0$ and $0 < t \leq 1$ whenever $d \geq A_{\eta'} \log(2n)$. With positive probability:*

- (S1) $\|Z\|_{\text{op}} \leq 16$, $\max_i \|Z^T v_i\|^2 \leq 3a$ and $\max_i \|Y_i\|^2 \leq 800a$;
- (S2) $\max_i |(\mathcal{A}Z)_i| \leq 3\sqrt{\rho a \log(2n)/n}$ and $\max_i |q_i(Z) - \mathbb{E}q_i(Z)| \leq \eta'a$;
- (S3) every $i \notin \mathfrak{B}$ has at least $0.95n$ indices $j \neq i$ with $|P_{ij} + tY_{ij}| \geq ht\sqrt{a/n}$;
- (S4) for every x supported on $\mathfrak{B}$,

$$|tx^T \mathcal{C}(Y)x| \leq \eta'(x^T Lx + at^2 \|x\|^2), \quad |x^T (\mathcal{L}(Y) - \mathbb{E}\mathcal{L}(Y))x| \leq \eta'a \|x\|^2.$$

Proof. Since $C_\rho \preceq I$, every linear functional $\langle Z, \Psi \rangle$ has variance at most $\|\Psi\|_{\text{F}}^2/n$; we use the tools of [Appendix A](#).

(S1): The net bound ([Lemma A.1\(c\)](#)) with $\sigma^2 = 1/n$ gives $\mathbb{P}(\|Z\|_{\text{op}} > 16) \leq 2 \cdot 5^n e^{-8n}$. The vector $Z^T v_i$ has covariance $\preceq n^{-1}I_d$ and trace $\leq a$; [Lemma A.1\(a\)](#) with $u = 2a$ (so $\min\{u^2/\|\cdot\|_{\text{F}}^2, u/\|\cdot\|_{\text{op}}\} \geq 2d$) and a union bound give $\max_i \|Z^T v_i\|^2 \leq 3a$ off probability $2ne^{-cd}$. Then $\|Y_i\| \leq \|Z^T v_i\| + \|Zu_i\| \leq \sqrt{3a} + 16\sqrt{3a/2}$.

(S2): $(\mathcal{A}Z)_i$ is Gaussian with variance $\leq \rho p_i/n \leq 3\rho a/(2n)$ ([Lemma 5.3\(b\)](#)); a union bound gives the first claim off probability $2n(2n)^{-3}$. The form $q_i(Z) = \langle Z, \mathbf{M}_i Z \rangle$, $\mathbf{M}_i Z = v_i v_i^T Z - Zu_i u_i^T$, has $\|\mathbf{M}_i\|_{\text{op}} \leq 1$ and $\|\mathbf{M}_i\|_{\text{F}}^2 \leq 2(d + np_i^2) \leq 7d$; after the covariance factor $(C_\rho/n)^{1/2}$ these become $\leq 1/n$ and $\leq \sqrt{7d}/n$, and [Lemma A.1\(a\)](#) with $u = \eta'a$ gives failure probability $2n \exp(-c\eta'^2 d)$.

(S3): *Variance bookkeeping.* By the proof of [Lemma 5.8](#), for $i \neq k$ with $P_{ik}^2 \leq a/16$ the unfiltered variance of Y_{ik} is at least $a/(8n)$; at most 24 indices k have $P_{ik}^2 > a/16$. Row i of T_{e_j} is $\delta_{ij}e_j^T P + P_{ij}e_j^T - 2P_{ij}e_j^T P$, of squared norm at most $3(\delta_{ij}p_j + P_{ij}^2 + 4P_{ij}^2 p_j)$, so the base part removes from row i total variance $\frac{1}{n} \sum_j \|e_j^T T_{e_j}\|^2 / p_j \leq \frac{6}{an} \cdot 5p_i \leq \frac{45}{n}$, which exceeds $\frac{a}{32n}$ at no more than $1440n/d$ positions. For $i \notin \mathfrak{B}$ the residual part removes at most $\delta_0 a$ from row i , hence more than $\frac{a}{32n}$ at no more than $32\delta_0 n$ positions. So for $d \geq 10^6$ (hence $n \geq 2 \cdot 10^6$, and the excluded indices number at most $25 + 0.0065n \leq 0.01n$) every $i \notin \mathfrak{B}$ has at least $0.99n$ indices $k \neq i$ with $\text{Var}(Y_{ik}) \geq \frac{a}{16n}$.

Small ball. Let $\phi : \mathbb{R} \rightarrow [0, 1]$ be C^1 , equal to 1 on $[-h, h]$, vanishing outside $[-2h, 2h]$, with $|\phi'| \leq 2/h$, and for $i \notin \mathfrak{B}$ let $F_i = \frac{1}{n} \sum_{k \neq i} \phi((P_{ik} + tY_{ik})/(t\sqrt{a/n}))$. By the density bound for a Gaussian of variance $\geq \frac{at^2}{16n}$, $\mathbb{E}F_i \leq 0.01 + \frac{4h}{\sqrt{2\pi/16}} \leq 0.01 + 7h$. As a function of the row $(Y_{ik})_k$, F_i has gradient norm at most $\frac{1}{n} \cdot \frac{2}{h} \cdot \sqrt{n/a} \cdot \sqrt{n} = 2/(h\sqrt{a})$; the row depends on Z through a linear map of norm ≤ 2 , and $Z = (C_\rho/n)^{1/2}g$ with g standard. So F_i is $4/(h\sqrt{d})$ -Lipschitz in g , and [Lemma A.1\(b\)](#) gives $\mathbb{P}(F_i > \mathbb{E}F_i + 0.01) \leq 2 \exp(-ch^2 d)$. Take $h = 1/500$, so that $\mathbb{E}F_i + 0.01 \leq 0.04$, and a union bound over i . On the complement, at most $0.04n$ indices $k \neq i$ have $|P_{ik} + tY_{ik}| < ht\sqrt{a/n}$, so at least $n - 1 - 0.04n \geq 0.95n$ have the reverse inequality.

(S4): Only second moments are needed, because $\mathfrak{B}$ is deterministic and bounded. Let $\Pi_{\mathfrak{B}}$ be the coordinate projection onto $\mathbb{R}^{\mathfrak{B}}$, $J_{\mathfrak{B}} = L_{\mathfrak{B}\mathfrak{B}} + at^2 I_{\mathfrak{B}}$ and $\omega_{ij} = J_{\mathfrak{B}}^{-1/2} \Pi_{\mathfrak{B}}(e_i - e_j)$, so $\|\omega_{ij}\|^2 \leq 2/(at^2)$ and $\sum_{i < j} P_{ij}^2 \omega_{ij} \omega_{ij}^T = J_{\mathfrak{B}}^{-1/2} L_{\mathfrak{B}\mathfrak{B}} J_{\mathfrak{B}}^{-1/2} \preceq I$. For x supported on $\mathfrak{B}$, $tx^T \mathcal{C}(Y)x = \langle J_{\mathfrak{B}}^{1/2} x, \Xi J_{\mathfrak{B}}^{1/2} x \rangle$ with $\Xi = \sum_{i < j} 2tP_{ij}Y_{ij} \omega_{ij} \omega_{ij}^T$, and $x^T J_{\mathfrak{B}} x = x^T Lx + at^2 \|x\|^2$. The coefficients $(Y_{ij})_{i < j}$ have covariance $\preceq \frac{2}{n}I$ (the map $Z \mapsto Y_Z$ is $\sqrt{2}$ times an isometry), so

$$\mathbb{E}\|\Xi\|_{\text{F}}^2 \leq \frac{2t^2}{n} \sum_{i < j} 4P_{ij}^2 \|\omega_{ij}\|^4 \leq \frac{16}{an} \text{tr}(J_{\mathfrak{B}}^{-1/2} L_{\mathfrak{B}\mathfrak{B}} J_{\mathfrak{B}}^{-1/2}) \leq \frac{16b}{d},$$

and Markov's inequality gives the first half off probability $16b/(d\eta'^2)$. The principal $\mathfrak{B}$ -block of $\mathcal{L}(Y) - \mathbb{E}\mathcal{L}(Y)$ has diagonal entries $\sum_{j \neq i} (Y_{ij}^2 - \mathbb{E}Y_{ij}^2)$, a centred Gaussian quadratic form of

variance at most $2 \cdot \frac{2}{n} \cdot \mathbb{E}\|Y_i\|^2 \leq \frac{12a}{n}$ (Lemma A.1(a), with $\mathbb{E}\|Y_i\|^2 \leq a(1-p_i) + (1-a)p_i \leq 3a$), and off-diagonal entries $-(Y_{ik}^2 - \mathbb{E}Y_{ik}^2)$ of variance at most $8/n^2$. Its expected squared Frobenius norm is at most $20b^2a/n = 20b^2a^2/d$, and Markov gives the second half off probability $20b^2/(d\eta'^2)$.

Finally choose $A_{\eta'}$ so large that $d \geq A_{\eta'} \log(2n)$ implies $d \geq 10^6$ and all failure probabilities above are at most $\frac{1}{10}$; their sum is less than 1. $\square$

5.6 The drifted retraction

Fix $D \geq 12$ and $t \in (0, 1]$, set $\rho = D^2t^2$, take a sample Z as in Lemma 5.9, and let m_* be as in Lemma 5.6. Define

$$W = Z + \frac{t}{2}m_*, \quad G = I + t^2W^TW, \quad U_t = (U + tVW)G^{-1/2}, \quad P_t = U_tU_t^T.$$

Since $U^TV = 0$, $(U + tVW)^T(U + tVW) = G$ and U_t is Parseval.

Lemma 5.10. *There is an absolute C_E (depending only on the bounds in (S1) and on $D \geq 12$) such that:*

- (a) $\|U_t - U\|_F^2 \leq C_E t^2 d$;
- (b) every row of $E = P_t - P - tY$ satisfies $\sum_j E_{ij}^2 \leq at^2(D^{-2} + C_E t^2)$;
- (c) with $\|r\|_\infty \leq C_E at^3$,

$$\text{diag } P_t - a\mathbf{1} = (1 - t^2)(p - a\mathbf{1}) + 2tAZ - t^2b_0 + t^2(q(Z) - \mathbb{E}q(Z)) + t^3q(Z, m_*) + \frac{t^4}{4}q(m_*) + r;$$

$$(d) \|\text{diag } P_t - a\mathbf{1}\|_\infty \leq at^2\left(\frac{\varepsilon}{t^2} + 6D\sqrt{\log(2n)/d} + \frac{12\varepsilon}{d} + \eta' + \frac{40}{D} + \frac{1}{2D^2} + C_E t\right).$$

Proof. By Lemma 5.6, $\frac{t}{2}\|m_*\|_{\text{op}} \leq \frac{t}{2}\|m_*\|_F \leq \sqrt{a/2}/D \leq \frac{1}{24}$, so with (S1): $\|W\|_{\text{op}} \leq 17$, $\|W\|_F \leq 17\sqrt{d}$, $\|W^Tv_i\| \leq 2\sqrt{a}$, $\|Wu_i\| \leq 17\sqrt{3a/2}$, and $\|(Y_{m_*})_i\|^2 \leq \|m_*\|_{\text{op}}^2 \leq 2a/\rho$. Hence $\|G^{-1} - I\|_{\text{op}}$ and $\|G^{-1/2} - I\|_{\text{op}}$ are at most $289t^2$, $\|U + tVW\|_{\text{op}} \leq 18$ and $\|u_i + tW^Tv_i\| \leq 4\sqrt{a}$.

$$(a) \|U_t - U\|_F \leq \|U + tVW\|_F \|G^{-1/2} - I\|_{\text{op}} + t\|W\|_F \leq Ct\sqrt{d}.$$

(b) $E = \frac{t^2}{2}Y_{m_*} + E'$ with $E' = P_t - P - tY_W = (U + tVW)(G^{-1} - I)(U + tVW)^T + t^2VWW^TV^T$, whose rows have norm at most $Ct^2\sqrt{a}$. The rows of $\frac{t^2}{2}Y_{m_*}$ have squared norm at most $\frac{t^4}{4} \cdot \frac{2a}{\rho} = \frac{at^2}{2D^2}$. Use $(\alpha + \beta)^2 \leq 2\alpha^2 + 2\beta^2$.

(c) Let $x_i = u_i + tW^Tv_i$, the i th row of $U + tVW$. Inserting $G^{-1} = I - t^2W^TW + t^4(W^TW)^2G^{-1}$ into $(P_t)_{ii} = x_i^TG^{-1}x_i$ and using $u_i^TW^Tv_i = (AW)_i$,

$$(P_t)_{ii} = p_i + 2t(AW)_i + t^2q_i(W) - 2t^3\langle Wu_i, WW^Tv_i \rangle - t^4\|WW^Tv_i\|^2 + t^4x_i^T(W^TW)^2G^{-1}x_i,$$

and the last three terms are $O(at^3)$. Now $2tAW = 2tAZ + t^2Am_*$, $q(W) = q(Z) + tq(Z, m_*) + \frac{t^2}{4}q(m_*)$, and $\mathbb{E}q(Z) = a\mathbf{1} - p - b_0 - Am_*$ by Lemma 5.6; the terms $\pm t^2Am_*$ cancel.

(d) In (c): $(1 - t^2)\|p - a\mathbf{1}\|_\infty \leq \varepsilon a$; $2t\|AZ\|_\infty \leq 6tDt\sqrt{a\log(2n)/n}$ by (S2), which is $6Dat^2\sqrt{\log(2n)/d}$; $t^2\|b_0\|_\infty \leq 12\varepsilon t^2/n$; the fluctuation is at most $\eta'at^2$; by (5.4) and (S1), $t^3|q_i(Z, m_*)| \leq t^3\sqrt{800a}\sqrt{2a/\rho} = 40at^2/D$ and $\frac{t^4}{4}|q_i(m_*)| \leq \frac{t^4}{4} \cdot \frac{2a}{\rho} = \frac{at^2}{2D^2}$. $\square$

Lemma 5.11 (Graph of the seed). *There are absolute $t_* \in (0, 1]$ and C_{core} such that the following holds. Let $\eta' \leq \frac{1}{32}$, $d \geq A_{\eta'} \log(2n)$, $D \geq 13/h$, $b \leq n/10$ and $t \leq t_*$, and let Z satisfy (S1)–(S4). Then $H(L_{P_t}) \leq C_{\text{core}}/(at^2)$.*

Proof. Core. In a row $i \notin \mathfrak{B}$, by Lemma 5.10(b), at most $at^2(D^{-2} + C_E t^2)/(\frac{h^2}{4} \frac{at^2}{n}) \leq 0.05n$ indices have $|\mathbf{E}_{ij}| \geq \frac{h}{2} t \sqrt{a/n}$, once $D \geq 13/h$ and $t^2 \leq h^2/(160C_E)$. With (S3), every $i \notin \mathfrak{B}$ has at least $0.9n$ indices $j \neq i$ with $(P_t)_{ij}^2 \geq \frac{h^2}{4} \frac{at^2}{n}$.

Block. For x supported on $\mathfrak{B}$, Lemma 5.8, $x^T K x \geq (n - b)\|x\|^2$ and (S4) give

$$x^T \mathcal{L}(P + tY)x \geq \left(1 - \eta' - t^2\left(\frac{2}{n} + \frac{8}{d}\right) - \frac{8}{D^2}\right)x^T Lx + at^2\left(\frac{1}{4}\left(1 - \frac{b}{n}\right) - 2\eta'\right)\|x\|^2 \geq \frac{at^2}{8}\|x\|^2,$$

since the first coefficient is positive and $\frac{1}{4} \cdot \frac{9}{10} - \frac{1}{16} \geq \frac{1}{8}$. By (5.1) and Lemma 5.10(b), $L_{P_t} = \mathcal{L}(P + tY + \mathbf{E}) \succeq \frac{1}{2}\mathcal{L}(P + tY) - 2at^2(D^{-2} + C_E t^2)I$, which is at least $\frac{at^2}{32}$ on vectors supported on $\mathfrak{B}$ once $t^2 \leq 1/(128C_E)$.

Barrier. Lemma 2.10 with $\mathcal{B} = \mathfrak{B}$, part (i) with $\vartheta = 0.4$, $\gamma = h^2 at^2/4$, and part (ii) with $\lambda = at^2/32$, gives $H(L_{P_t}) \leq \frac{1+b}{0.1h^2 at^2} + \frac{32\sqrt{b}}{at^2}$. Put $C_{\text{core}} = 10(1 + b_{\max})/h^2 + 32\sqrt{b_{\max}}$. $\square$

5.7 Proof of Theorem 5.1

Constants. The numbers $\delta_0, b_{\max}, h, C_E, t_*, C_{\text{core}}$ are absolute and do not depend on D (for $D \geq 12$). Put $\Theta = \max\{2, C_{\text{core}}, C_E\}$ and choose, in this order: $D \geq \max\{13/h, 500\Theta\}$, so that $\frac{40}{D} + \frac{1}{2D^2} \leq \frac{1}{12\Theta}$; $K_0 = 12\Theta$ and $\eta' = \min\{\frac{1}{32}, \frac{1}{12\Theta}\}$; $A \geq A_{\eta'}$ so large that $d \geq A \log(2n)$ implies $6D\sqrt{\log(2n)/d} \leq \frac{1}{12\Theta}$ and $b_{\max} \leq n/10$; and $\varepsilon_0 \leq \frac{1}{2}$ so that $t = \sqrt{K_0\varepsilon}$ satisfies $t \leq t_*$, $C_E t \leq \frac{1}{12\Theta}$ and $12\varepsilon \leq \frac{1}{12\Theta}$ for $\varepsilon \leq \varepsilon_0$. None of these choices depends on n, d, ε or U .

Seed. With $t^2 = K_0\varepsilon$, Lemma 5.10(d) bounds the diagonal error of P_t by $at^2 \cdot \frac{6}{12\Theta} = \frac{at^2}{2\Theta}$; Lemma 5.11 bounds the barrier constant by $\Theta/(at^2)$; and Lemma 5.10(a) bounds $\|U_t - U\|_{\text{F}}^2$ by $\Theta t^2 d$. So U_t is a (Θ, t) -seed for U , and Corollary 2.13 gives an ENP frame $\hat{U}$ with $\|\hat{U} - U\|_{\text{F}}^2 \leq 3\Theta t^2 d = 36\Theta^2 \varepsilon d$. $\square$

Remark 5.12. Without the drift one would have to remove the bias $t^2 b_*$ after the fact, for instance by a second scaling; since $\langle x, b_* \rangle^2 \leq \frac{2a}{\rho} x^T L x$ this is possible, but it requires two-sided control of L_{P_t} on all of $\mathbf{1}^\perp$. With the drift, the only spectral input is the gap on the bounded deterministic block $\mathfrak{B}$, which needs second moments only.

6 Assembly

We use the two seeds through the following equal-norm formulation.

Proposition 6.1. *There is an absolute constant C_0 such that every $X \in \mathbb{R}^{n \times d}$ with $1 \leq d$, $2d \leq n$ and $\|x_i\|^2 = a$ for all i admits an ENP frame W with*

$$\|X - W\|_{\text{F}}^2 \leq C_0 d \|X^T X - I\|_{\text{op}}.$$

Proof. Let $\eta = \|X^T X - I\|_{\text{op}}$; if $\eta = 0$ take $W = X$. Let A, ε_0, C_m be the constants of Theorem 5.1 and B, c_*, C_* those of Theorem 4.1, and fix an integer $d_0 \geq 2$ with $A \log(2Bd^2) \leq d$ for all $d \geq d_0$.

Case $d < d_0$. Lemma 3.1 gives $\|X - W\|_{\text{F}}^2 \leq \eta d(d-1) \leq d_0 \eta d$.

Case $d \geq d_0$, $n \geq Bd^2$. If $\eta \leq c_*$ use Theorem 4.1; otherwise Corollary 3.2 gives cost $4d \leq (4/c_*)\eta d$.

Case $d \geq d_0$, $n < Bd^2$. Then $A \log(2n) \leq d$. If $\eta \leq \varepsilon_0/4$, let $U = X(X^T X)^{-1/2}$. The singular values of X lie in $[\sqrt{1-\eta}, \sqrt{1+\eta}]$, so $\|X - U\|_{\text{F}}^2 \leq d\eta^2$, and $\|u_i\|^2 = x_i^T (X^T X)^{-1} x_i \in [a/(1+\eta), a/(1-\eta)]$, so $|\|u_i\|^2 - a| \leq 2\eta a$. Theorem 5.1 with $\varepsilon = 2\eta$ gives an ENP W with $\|U - W\|_{\text{F}}^2 \leq 2C_m \eta d$, and $\|X - W\|_{\text{F}}^2 \leq 2d\eta^2 + 4C_m \eta d$. If $\eta > \varepsilon_0/4$ use the trivial cost $4d \leq (16/\varepsilon_0)\eta d$. Altogether $C_0 = \max\{d_0, C_*, 4/c_*, 2 + 4C_m, 16/\varepsilon_0\}$ works. $\square$

Proof of Theorem 1.2. Replacing (P, Q) by $(I - P, I - Q)$ preserves β , ranks of complements and distances, so we may assume $d \leq n/2$; the case $d = 0$ is trivial. Write $P = UU^T$ with $U^T U = I$, $p_i = \|u_i\|^2$ and $\delta = \beta/a$. If $\delta \geq \frac{1}{2}$, any ENP W (Corollary 3.2) gives $\|P - WW^T\|_F^2 \leq 2d \leq 4n\beta$. Otherwise let $x_i = \sqrt{a}u_i/\|u_i\|$. Then $\|X - U\|_F^2 = \sum_i (\sqrt{p_i} - \sqrt{a})^2 \leq \sum_i (p_i - a)^2/a \leq \delta^2 d$ and $X^T X = U^T \text{Diag}(a/p)U$, so $\|X^T X - I\|_{\text{op}} \leq \max_i |a/p_i - 1| \leq \delta/(1 - \delta) \leq 2\delta$. Proposition 6.1 gives an ENP W with $\|U - W\|_F^2 \leq 2\delta^2 d + 4C_0\delta d \leq (1 + 4C_0)n\beta$, using $\delta d = n\beta$. Take $Q = WW^T$; by Lemma 2.1, $\|P - Q\|_F^2 \leq 2(1 + 4C_0)n\beta$. So Theorem 1.2 holds with $C_P = \max\{4, 2 + 8C_0\}$. $\square$

Proof of Theorem 1.1. If $\varepsilon \geq \frac{1}{2}$, an ENP W gives $\|X - W\|_F^2 \leq 2\|X\|_F^2 + 2d \leq 6d \leq 12\varepsilon d$, since $\|X\|_F^2 = \text{tr } X^T X \leq (1 + \varepsilon)d$. Let $\varepsilon < \frac{1}{2}$ and $U = X(X^T X)^{-1/2}$, $P = UU^T$. As above $\|X - U\|_F^2 \leq \varepsilon^2 d$ and $P_{ii} = x_i^T (X^T X)^{-1} x_i \in [\frac{1-\varepsilon}{1+\varepsilon}a, \frac{1+\varepsilon}{1-\varepsilon}a]$, so $\beta = \max_i |P_{ii} - a| \leq 4\varepsilon a$. Theorem 1.2 gives Q with $\|P - Q\|_F^2 \leq 4C_P \varepsilon d$, and by Lemma 2.1 there is an isometry W spanning Q —hence ENP—with $\|U - W\|_F^2 \leq \|P - Q\|_F^2$. Then $\|X - W\|_F^2 \leq 2\varepsilon^2 d + 8C_P \varepsilon d$, so Theorem 1.1 holds with $C = \max\{12, 1 + 8C_P\}$. $\square$

A Gaussian tools

Lemma A.1. *The following hold with absolute constants.*

(a) (Quadratic forms.) *For g standard Gaussian and A symmetric, $\text{Var}(g^T A g) = 2\|A\|_F^2$ and*

$$\mathbb{P}\{|g^T A g - \text{tr } A| > u\} \leq 2 \exp\left[-\frac{1}{8} \min\left\{\frac{u^2}{\|A\|_F^2}, \frac{u}{\|A\|_{\text{op}}}\right\}\right].$$

For a Gaussian vector $\zeta = \Sigma^{1/2}g$, apply this to $\Sigma^{1/2}A\Sigma^{1/2}$, whose norms are at most $\|\Sigma\|_{\text{op}}$ times those of A .

(b) (Lipschitz functions.) *If F is C^1 with $\|\nabla F\| \leq \ell$ and g is standard Gaussian, then $\mathbb{P}\{|F(g) - \mathbb{E}F(g)| > u\} \leq 2 \exp(-2u^2/(\pi^2 \ell^2))$.*

(c) (Nets.) *If Z is an $r \times s$ centred Gaussian matrix with $\text{Cov}(\text{vec } Z) \preceq \sigma^2 I$, then $\mathbb{P}\{\|Z\|_{\text{op}} > u\} \leq 2 \cdot 5^{r+s} \exp(-u^2/(32\sigma^2))$.*

Proof. (a) Diagonalise A with eigenvalues λ_j . For $|s| \leq 1/(4\|A\|_{\text{op}})$, $\log \mathbb{E}e^{s(g^T A g - \text{tr } A)} = \sum_j [-s\lambda_j - \frac{1}{2} \log(1 - 2s\lambda_j)] \leq 2s^2\|A\|_F^2$, since $-x - \frac{1}{2} \log(1 - 2x) = \sum_{k \geq 2} (2x)^k/(2k) \leq 2x^2$ for $|x| \leq \frac{1}{4}$. The exponential Markov inequality with $s = \min\{u/(4\|A\|_F^2), 1/(4\|A\|_{\text{op}})\}$, applied to both signs, gives the tail. The variance follows from $\mathbb{E}g_j^4 = 3$.

(b) Let g' be an independent copy, $g_\theta = g \cos \theta + g' \sin \theta$ and $\dot{g}_\theta = -g \sin \theta + g' \cos \theta$; for each θ the pair $(g_\theta, \dot{g}_\theta)$ is standard Gaussian. By the fundamental theorem of calculus and Jensen on $[0, \pi/2]$,

$$\mathbb{E}e^{s(F(g') - F(g))} \leq \frac{2}{\pi} \int_0^{\pi/2} \mathbb{E} \exp\left[\frac{\pi s}{2} \langle \nabla F(g_\theta), \dot{g}_\theta \rangle\right] d\theta \leq e^{\pi^2 s^2 \ell^2 / 8}.$$

Jensen in g' gives $\mathbb{E}e^{s(F(g') - \mathbb{E}F)} \leq e^{\pi^2 s^2 \ell^2 / 8}$; optimise in s . All expectations are finite since $|F(x)| \leq |F(0)| + \ell\|x\|$, and Fubini applies to the θ -integral.

(c) Let $\mathcal{N}_1, \mathcal{N}_2$ be $\frac{1}{2}$ -nets of the unit spheres of $\mathbb{R}^r$ and $\mathbb{R}^s$ with at most $5^r, 5^s$ points. For a vector w , a net point y' within $\frac{1}{2}$ of $w/\|w\|$ has $\langle w, y' \rangle \geq \frac{1}{2}\|w\|$; applying this twice gives $\|Z\|_{\text{op}} \leq 4 \max_{x \in \mathcal{N}_1, y \in \mathcal{N}_2} |x^T Z y|$. Each $x^T Z y$ is Gaussian with variance $\leq \sigma^2$; a union bound finishes. $\square$

Covariance domination. If a Gaussian vector ξ has covariance $\Gamma \preceq \gamma I$ and (M_α) are matrices, then $\mathbb{E}\|\sum_\alpha \xi_\alpha M_\alpha\|_F^2 = \text{tr}(\Gamma G) \leq \gamma \sum_\alpha \|M_\alpha\|_F^2$, where $G \succeq 0$ is the Gram matrix of the M_α . Correlated Gaussians enter only as $\Sigma^{1/2}g$: in quadratic forms through the last sentence of (a) (Lemma 4.3, Lemma 5.9), in Lipschitz functions through (b), in operator norms through (c), and in Frobenius norms of linear combinations through this identity (Lemma 5.9(S4)).

B Formalisation

Theorem 1.1 is formalised in Lean 4 with Mathlib; the development accompanies this paper (directory `lean/`). The formal statement is

```
def SharpPaulsenBound : Prop :=
  ∃ C : ℝ, 0 < C ∧
    ∀ (n d : ℕ), 0 < d → d ≤ n →
      ∀ (ε : ℝ) (U : Frame n d), 0 < ε → ε < 1 →
        IsNearlyEqualNormParseval ε U →
          ∃ W : Frame n d, IsEqualNormParseval W ∧
            sqDistance U W ≤ C * ε * (d : ℝ)
```

where `Frame n d` is the type of real $n \times d$ matrices, Parsevalness is $U^T U = 1$, the near-Parseval condition is stated through the quadratic form $x \mapsto \|Ux\|^2$, the row-norm conditions refer to d/n as a real quotient, and `sqDistance` is the unnormalised squared Frobenius distance. The projection form (Theorem 1.2) is derived for arbitrary real symmetric idempotent matrices of rank d . Both statements are also restated in plain Mathlib vocabulary (positive semidefiniteness, dot products, traces, `Matrix.rank`), and the restatements are derived from them.

Structure. The development has about 29,600 lines. A library of about 26,000 lines contains frames, projections and alignment, the squared-Gram Laplacian, diagonal scaling and the log-determinant potential, complementation and normalisation, the Hamilton–Moitra argument (Section 3), Gaussian tools (quadratic forms, Lipschitz functions, nets, covariance calculus), the constrained and row-independent noises and the row-normalisation identity of Section 4, and the filtered covariance, the mean formula and the sampling estimates of Section 5. About 3,600 lines (directory `Paulsen/Linear`) implement the argument specific to this paper:

paper	Lean (Paulsen/Linear/)
Definition 2.6 and Lemma 2.7	Barrier: HasBarrierBound, barrier_subset_bound
Theorem 2.8	Balancing: balancing_cost_barrier
Lemma 2.10 (main part, (i), (iii))	Core, ManyRowBarrier
Lemma 2.10(ii)	BlockBarrier: block_barrier
Definition 2.12 and Corollary 2.13	Seed: IsSeed.hasCorrection
Lemma 4.2	ManyRowMoments
Lemma 4.4	ManyRowRemainder: tangentRemainder_centred_bound
Theorem 4.1	ManyRowGraph, ManyRowSample, ManyRowParameters, ManyRow
Lemma 5.5(a), Lemma 5.6 (m_*), (5.4)	DriftAlgebra: driftGamma, driftMean
Lemma 5.10	DriftRetraction, DriftDiagonal
Lemma 5.11, Theorem 5.1	ModerateBarrier, ModerateDrift
Proposition 6.1, Theorems 1.1 and 1.2	Assembly, Main

Differences from the text. The formal proof follows the argument above, with these differences of detail.

- (a) All constants are much larger (for instance $B = 10^{40}$, $c_* = 10^{-60}$ in [Theorem 4.1](#); $A = 10^{130}$, $\varepsilon_0 = 10^{-130}$, $D = 10^{40}$ in [Theorem 5.1](#)); the balancing cost is $8\|q - p\|_1$ rather than $\frac{15}{4}\|q - p\|_1$; the net bound is used with $\|Z\|_{\text{op}} \leq 128$; and [Lemma 2.10\(ii\)](#) is used with $R = b/\lambda$.
- (b) In [Lemma 5.9](#), the block estimate (S4) is not proved from second moments. Instead the formalisation uses two global events that are also available from the library when $d \geq A \log 2n$: the concentration $\|\mathcal{L}(Y) - \mathbb{E}\mathcal{L}(Y)\|_{\text{op}} \leq a/32$ of the squared Gaussian graph, and the bound $|t x^T \mathcal{C}(Y) x| \leq \frac{1}{32}(x^T L x + a t^2 \|x\|^2)$ for all x . Together with [Lemma 5.8](#) they give a gap $\frac{a t^2}{32}$ for L_{P_t} on $\mathbf{1}^\perp$, and subtracting the mean gives the gap on vectors supported on $\mathfrak{B}$ that [Lemma 5.11](#) needs. The rest of the moderate-row argument is as in the text; in particular, nothing is corrected after the drifted retraction.
- (c) In [Lemma 4.2](#) the fourth moment of a centred Gaussian quadratic form is bounded through its moment generating function, which gives a worse constant. One functional, $a \sum_i (r_i^2 - 1)^2$, controls both $\|\Gamma Z\|_{\text{F}}^2$ and $\|\Lambda X\|_{\text{F}}^2$.
- (d) [Lemma 4.6](#) is used in a slightly different form: the dense reference graph has $\frac{17}{20}n$ heavy neighbours at every row, which holds for every $d \geq 1$.

Checks. The project builds with Lean 4.32.0 and Mathlib v4.32.0. `Audit.lean` checks that the final theorems have the target types; that every theorem of the project depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`; and that the proof of the main theorem uses the barrier toolbox, the centred remainder, the drift and the partition-free assembly, and none of the Poisson-constant endpoints of the library. Every module was also replayed in the kernel with `leanchecker`.

Contributions and acknowledgements

K. Hänni and H. Eberhard proposed the overall strategy: perturb by a random tangent of size $\sqrt{\varepsilon}$ to create a well-connected squared-Gram graph, and then balance by a diagonal scaling controlled in ℓ_∞ . Astra developed the first complete proof along these lines, including both seeds and the reduction to them, and formalised it in Lean. Claude found the simplifications in the present version (the cancelling drift in the moderate-row seed, the centred remainder estimate that makes the many-row seed work at every error level and removes a block-partition step, and the barrier-constant formulation of balancing), wrote this text, and formalised the present proof.

References

- [1] J. Cahill and P. G. Casazza. *The Paulsen problem in operator theory*. Operators and Matrices 7 (2013). [arXiv:1102.2344](#).
- [2] L. Hamilton and A. Moitra. *The Paulsen problem made simple*. ITCS 2019; Israel J. Math. (2021). [arXiv:1809.04726](#).
- [3] T. C. Kwok, L. C. Lau, Y. T. Lee and A. Ramachandran. *The Paulsen problem, continuous operator scaling, and smoothed analysis*. STOC 2018. [arXiv:1710.02587](#).
- [4] L. C. Lau and A. Ramachandran. *Optimal bounds for Tyler’s M-estimator for elliptical distributions*, Section 7. [arXiv:2510.13751](#), 2025.